

PARIKSHA365 — STUDY NOTES

Chemistry — Common Book

SSC, RRB, State PSC, Banks (GA) — General Science

SSC

RRB

BANKS

STATE PSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

Examiner's Blueprint — What Chemistry Questions Are Really About

EXAM ALERT

Chemistry questions are almost never about deep understanding — they are about specific, well-targeted recall.

Three categories account for 80% of chemistry questions:

- **1. Everyday names vs chemical names** — "Chemical name of baking soda?" Appears in almost every paper.
- **2. Alloy composition** — "Which metals make brass?" Expect at least one in every General Science section.
- **3. Discovery / inventor facts** — "Who invented nylon?" Steady 1–2 questions per paper in SSC exams.

Everything else — pH values, activity series, industrial processes, polymers, indicators — shows up regularly but less predictably.

This book is built around the three hot zones, then covers the rest in depth so you're never caught off guard.

Priority order for your study time:

1. Everyday chemical names table (Part U) — memorise every row
2. Alloys table (Part V) — know composition + use for top 15 alloys
3. Discovery/inventor dates (Part T) — the "who discovered X" questions
4. Acids, bases, pH, indicators (Part C-D) — 2-3 Q every paper
5. Activity series + metal reactions (Part E) — displacement questions
6. Polymers + their monomers (Part F) — SSC loves this
7. Industrial processes — Haber, Contact, Hall-Héroult (Part AA)
8. States of matter + periodic table trends (Parts A-B)
9. Organic basics — functional groups, fuels (Parts G-H)
10. Electrochemistry — Faraday's laws, batteries, galvanic cells (Part J)
11. Fertilisers, food chemistry, environmental (Parts Y, I)

How to read this book — your real study load

Total pages: 57. Don't let the page count scare you.

Mandatory reading — pages 1 to 37 (65% of the book)

That's the full syllabus. Master those 37 pages and you've covered every concept an examiner can fairly ask.

Bonus material — pages 38 to 57 (35% of the book)

Reference tables, compilations, and self-test material — useful in your last week of revision OR if you're aiming for a top rank.

Two study tracks

- **Just want to pass?** Read pages 1–37. Skim the appendices once before the exam.
 - **Want to top your batch?** Read the appendices too — they're how 70% becomes 90%.
-

Index — Table of Contents (clickable)

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p21–23	PART E	METALS AND THE ACTIVITY SERIES
p24–28	PART F	ORGANIC CHEMISTRY
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p32–34	PART H	EVERYDAY CHEMISTRY
p35–37	PART I	ENVIRONMENTAL CHEMISTRY
p38–41	PART J	ELECTROCHEMISTRY — <i>Electrolysis, Faraday's laws, galvanic cells, batteries, electrode potentials.</i>
p42–43	PART T	DISCOVERIES AND INVENTORS IN CHEMISTRY — <i>Discoveries & Inventors — who-discovered-what table.</i>
p40–41	PART U	THE ESSENTIAL CHEMICAL NAMES TABLE — <i>Essential Chemical Names — everyday ↔ chemical name table.</i>
p42–43	PART V	ALLOYS AND METALLURGY — <i>Alloys & Metallurgy — composition tables.</i>
p44–46	PART Y	FERTILISERS — <i>Fertilisers — N/P/K compositions, NPK ratios.</i>
p47–48	PART AA	INDUSTRIAL PROCESSES — <i>Industrial Processes — Haber, Contact, Hall-Hérault, Solvay.</i>

Pages	Part / Chapter	Topic / What it covers
p49–52	PART AB	ORGANIC CHEMISTRY DEPTH — <i>Organic Chemistry Depth</i> — extra worked detail.
p53–57	PART AC	BIOMOLECULES — <i>Biomolecules</i> — carbs, proteins, lipids, nucleic acids.

PART A — MATTER AND ITS NATURE

Chapter A1 — States of Matter

The intuition — ice on a hot day

- Picture a ₹10 ice block on a summer day.
- In 20 minutes: **solid** → **liquid** → **gas** (steam).
- Same H₂O molecules throughout — only motion and spacing change.

That is the entire story of states of matter.

The Five States

- **Solid** — definite shape and volume; particles vibrate in fixed positions around a lattice point.
- **Liquid** — definite volume, no fixed shape; particles slide over each other.
- **Gas** — no fixed shape or volume; particles move freely and fill any container.
- **Plasma** — ionised gas where electrons are stripped from atoms. Found in the Sun, stars, lightning, fluorescent tubes, neon signs.
- **Bose-Einstein Condensate (BEC)** — occurs at temperatures near absolute zero (−273 °C). Particles lose individual identities and behave as a single "super-atom." Predicted by Satyendra Nath Bose and Albert Einstein; first created in a lab in 1995.

KEY FACT

The 5th state of matter is Bose-Einstein Condensate — NOT plasma. Examiners specifically ask this. Plasma is common around us (the Sun, tube lights, lightning); BEC is exotic and lab-made.

Phase Change Names (all six)

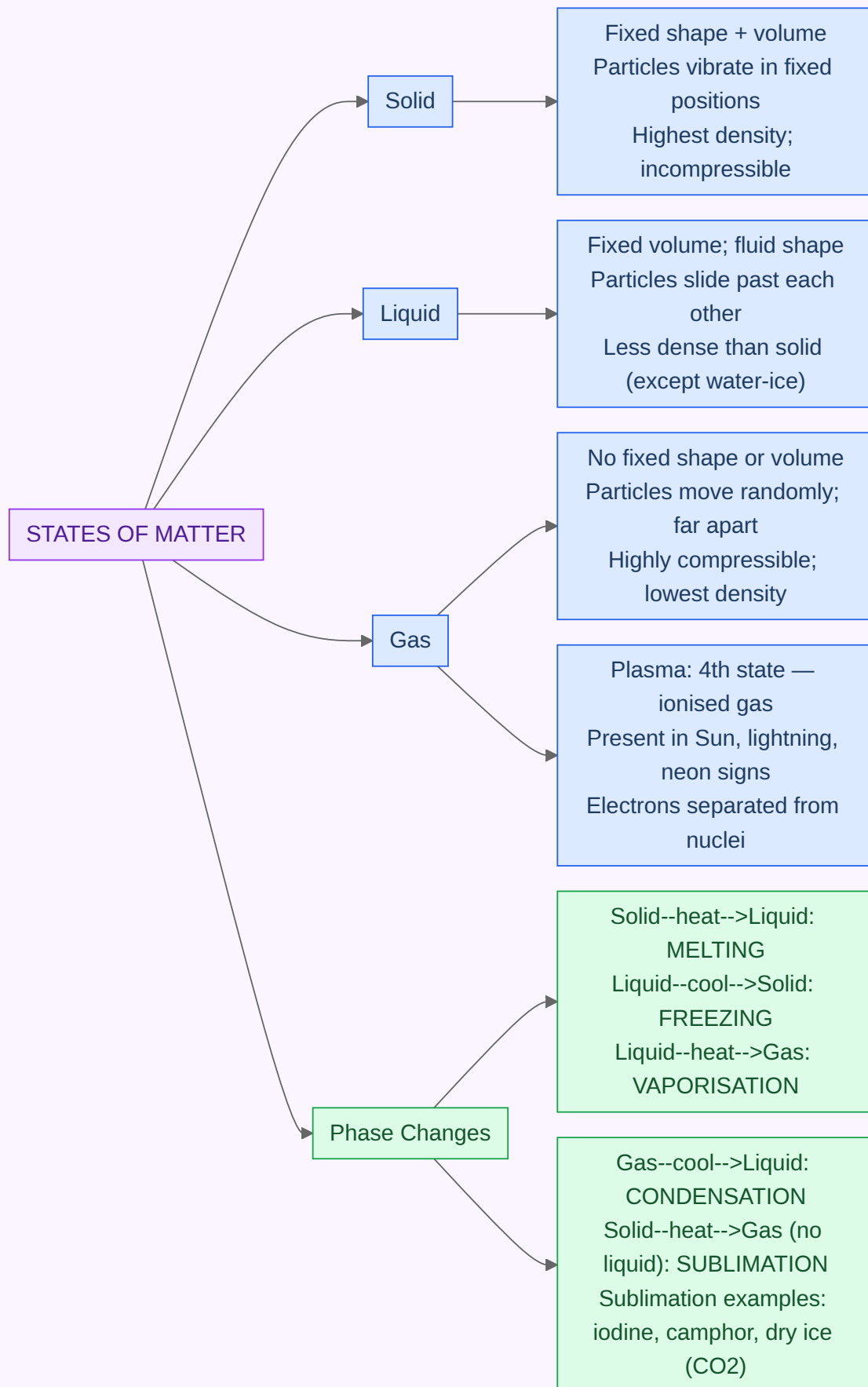
From	To	Name
Solid	Liquid	Melting (Fusion)
Liquid	Solid	Freezing (Solidification)
Liquid	Gas	Evaporation / Boiling
Gas	Liquid	Condensation
Solid	Gas	Sublimation
Gas	Solid	Deposition (reverse sublimation)

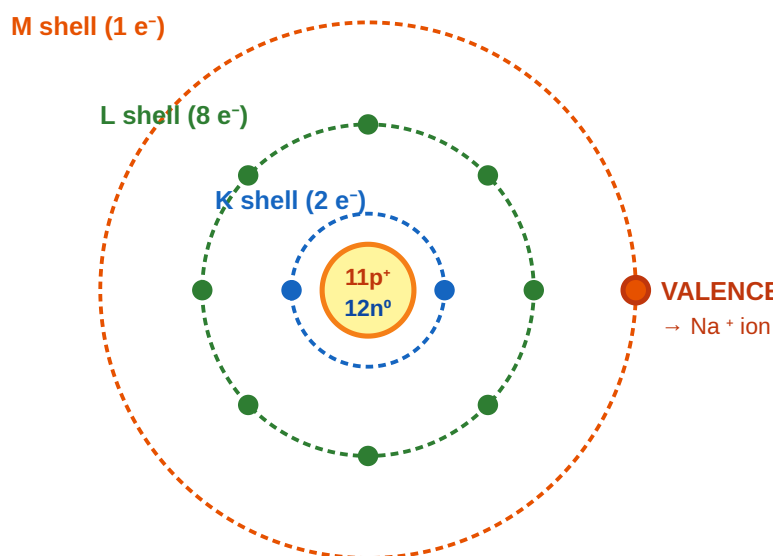
EXAM ALERT

Sublimation is a high-frequency topic. Substances that sublime directly from solid to gas without becoming liquid: **dry ice (CO₂)**, **iodine**, **naphthalene (moth balls)**, **camphor**, **ammonium chloride**. Frost

forming on a cold window is deposition (reverse sublimation).

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Bohr Model — Atom Structure (Sodium, $Z=11$)

Atom Basics

- Z = atomic number
= number of protons
- A = mass number
= protons + neutrons
- Neutrons = $A - Z$

Shell capacity

- $K = 2$ ($n=1$)
- $L = 8$ ($n=2$)
- $M = 18$ ($n=3$)
- $N = 32$ ($n=4$)
- ($2n^2$ rule)

Sub-atomic mass

- Proton ≈ 1 u (+1)
- Neutron ≈ 1 u (0)
- Electron $\approx 1/1836$ u
- Heaviest = NEUTRON

The Atomic Model: A History Worth Knowing

"Who discovered what?" — appears 1–2 times per paper. Learn this sequence cold.

Scientist	Year	Contribution
John Dalton	1808	Atoms are indivisible; each element has characteristic atoms
J.J. Thomson	1897	Discovered the electron ; proposed plum-pudding model
Ernest Rutherford	1911	Alpha-scattering experiment → nucleus discovered; "mostly empty space"
Henry Moseley	1913	Measured atomic number using X-rays; fixed the periodic table order
Niels Bohr	1913	Quantised electron orbits; explained hydrogen spectrum
James Chadwick	1932	Discovered the neutron
Quantum model	1926+	Electrons occupy probability clouds (orbitals), not fixed paths

KEY FACT

Three discoveries that examiners repeat: **electron** → J.J. Thomson (1897), **proton** → Rutherford (1919), **neutron** → Chadwick (1932). The heaviest sub-atomic particle is the neutron (just slightly heavier than the proton).

Key Definitions

- **Atomic number (Z)** = number of protons = number of electrons in a neutral atom.
- **Mass number (A)** = protons + neutrons.
- **Isotopes** — same Z , different A . Example: Hydrogen has three isotopes: ^1H (protium), ^2H (deuterium), ^3H (tritium). Heavy water uses deuterium: D_2O .

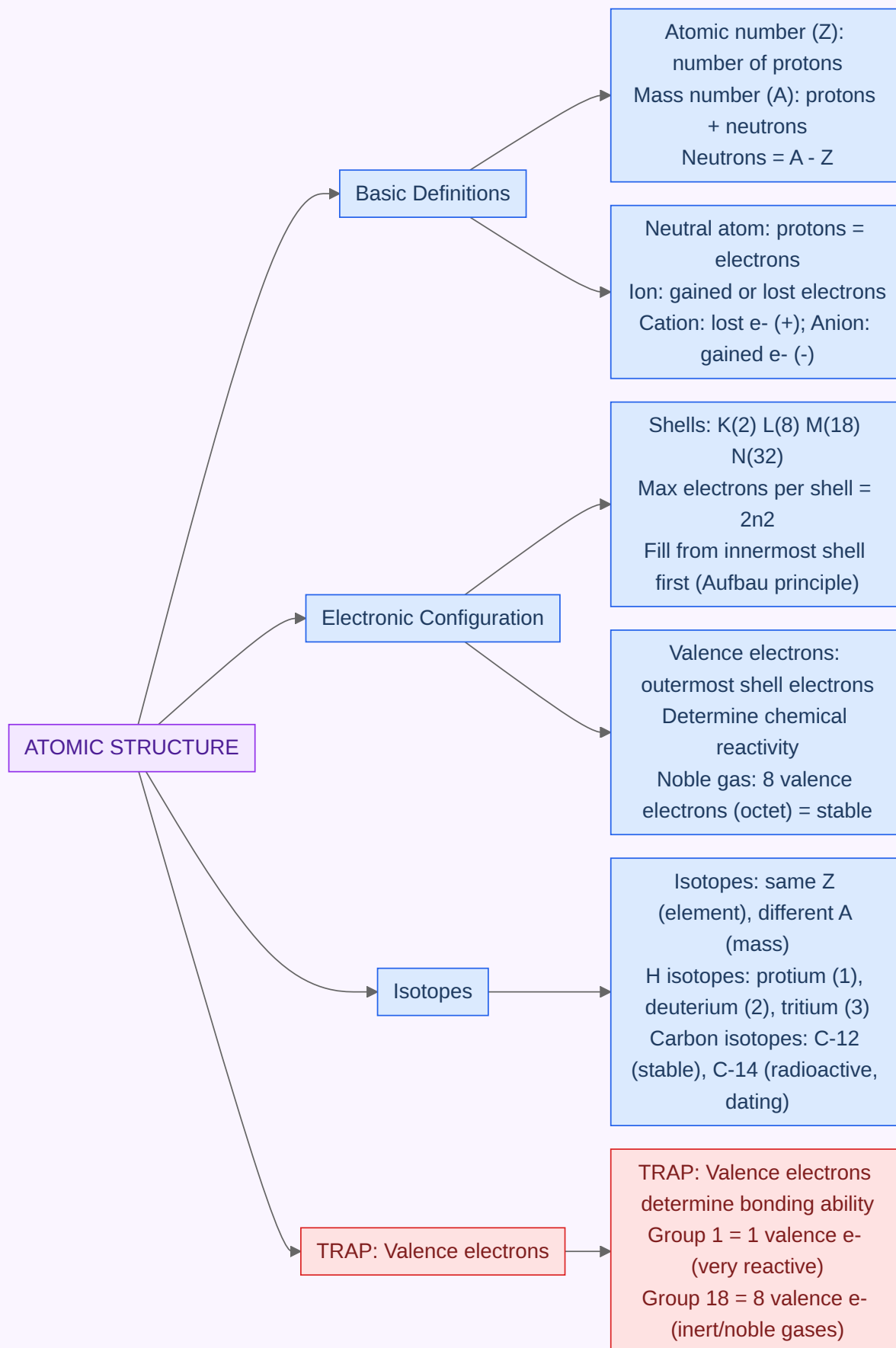
- **Isobars** — same A, different Z. Example: ^{40}Ar , ^{40}K , ^{40}Ca .
- **Isotones** — same number of neutrons, different Z.

REMEMBER

To remember isotopes vs isobars: "**Same Z = IsoTopes (T = Top = atomic number on top in notation)**" and "**Same A = IsobArs (A = same mass number)**".

PART B — THE PERIODIC TABLE

⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter B1 — From Mendeleev to Moseley

Mendeleev's table (1869)

- **Dmitri Mendeleev** arranged 63 known elements by **atomic weight**.
- Noticed repeating patterns in behaviour → called it the **periodic law**.
- **Left gaps** for elements that didn't yet exist.
- Predicted: **Eka-aluminium** (= Gallium), **Eka-silicon** (= Germanium), **Eka-boron** (= Scandium).
- All three were discovered within 15 years, confirming the table.

The Mendeleev problem

- Argon (Ar, 39.9) came **before** potassium (K, 39.1) by weight — but Ar clearly belonged with noble gases.
- Co–Ni and Te–I were similarly reversed.

Moseley's fix (1913)

- Re-ordered by **atomic number** (= number of protons), not atomic weight.
- All anomalies vanished.
- The **modern periodic law** is based on atomic number.

KEY FACT

The modern periodic table has **7 periods (rows)**, **18 groups (columns)**, **118 elements**, and is divided into **4 blocks: s, p, d, f**. The last four elements (113, 115, 117, 118 — Nihonium, Moscovium, Tennessine, Oganesson) were officially added in **2016**.

The Four Blocks

- **s-block** (Groups 1–2) — Alkali + alkaline-earth metals. Li, Na, K | Be, Mg, Ca. Soft, reactive, low MP.
- **p-block** (Groups 13–18) — The most variety: metals + metalloids + non-metals + noble gases.
- **d-block** (Groups 3–12) — Transition metals. Fe, Cu, Au, Ag, Pt. Hard, high MP, variable oxidation states.
- **f-block** (below main table) — Lanthanides (4f, rare earths) + Actinides (5f, radioactive). All elements beyond U are man-made.

Periodic Table — 4 Blocks (s, p, d, f)

H S Gr 1–2 Alkali + Alkaline earth Li Na K Mg Ca Soft, reactive Low MP	d Gr 3–12 TRANSITION METALS Fe Cu Ag Au Zn Pt Ni Cr Mn Ti Hard, high MP Variable oxidation states	p He Gr 13–18 Metals + Metalloids + Non-metals + Noble gases C N O F Ne Al Si P S Cl The most variety
Lanthanides (4f): La–Lu, "rare earths" Actinides (5f): Ac–Lr, all radioactive (incl. U, Pu) Beyond Uranium = man-made (transuranic)		

7 periods • 18 groups • 118 elements (last 4 added 2016: Nh, Mc, Ts, Og)

Important Groups — Learn These Families

Group	Family	Members	Character
1	Alkali metals	Li, Na, K, Rb, Cs, Fr	1 valence e ⁻ ; very reactive; soft
2	Alkaline earth metals	Be, Mg, Ca, Sr, Ba, Ra	2 valence e ⁻ ; less reactive than Group 1
13	Boron family	B, Al, Ga, In, Tl	3 valence e ⁻
14	Carbon family	C, Si, Ge, Sn, Pb	4 valence e ⁻
15	Pnictogens	N, P, As, Sb, Bi	5 valence e ⁻
16	Chalcogens	O, S, Se, Te, Po	6 valence e ⁻
17	Halogens	F, Cl, Br, I, At	7 valence e ⁻ ; most reactive non-metals
18	Noble gases	He, Ne, Ar, Kr, Xe, Rn	Full outer shell; inert

REMEMBER

Noble gases: "**He Never Argues; Keeps Xtra Rest**" — He, Ne, Ar, Kr, Xe, Rn. Halogens: "**Fierce Clowns Bring Interesting Acts**" — F, Cl, Br, I, At.

Periodic Trends — The Direction Rules

Trend	Across a period (left → right)	Down a group
Atomic size	Decreases	Increases
Ionisation energy	Increases	Decreases
Electron affinity	Increases	Decreases
Electronegativity	Increases	Decreases
Metallic character	Decreases	Increases

KEY FACT

Quick Recall — Periodic Trends: Moving RIGHT across a period: atoms shrink (more protons pull electrons in), harder to remove electrons (higher IE), more electronegative. Moving DOWN a group: atoms grow (new shells), easier to lose electrons, more metallic. Think: "**Right = tighter; Down = looser.**"

REMEMBER

Trend direction trick: Atomic size is the **opposite** of all other trends. Size DECREASES going right (others increase). Size INCREASES going down (others decrease). So just learn size direction and flip it for IE/EN/EA.

EXAM ALERT

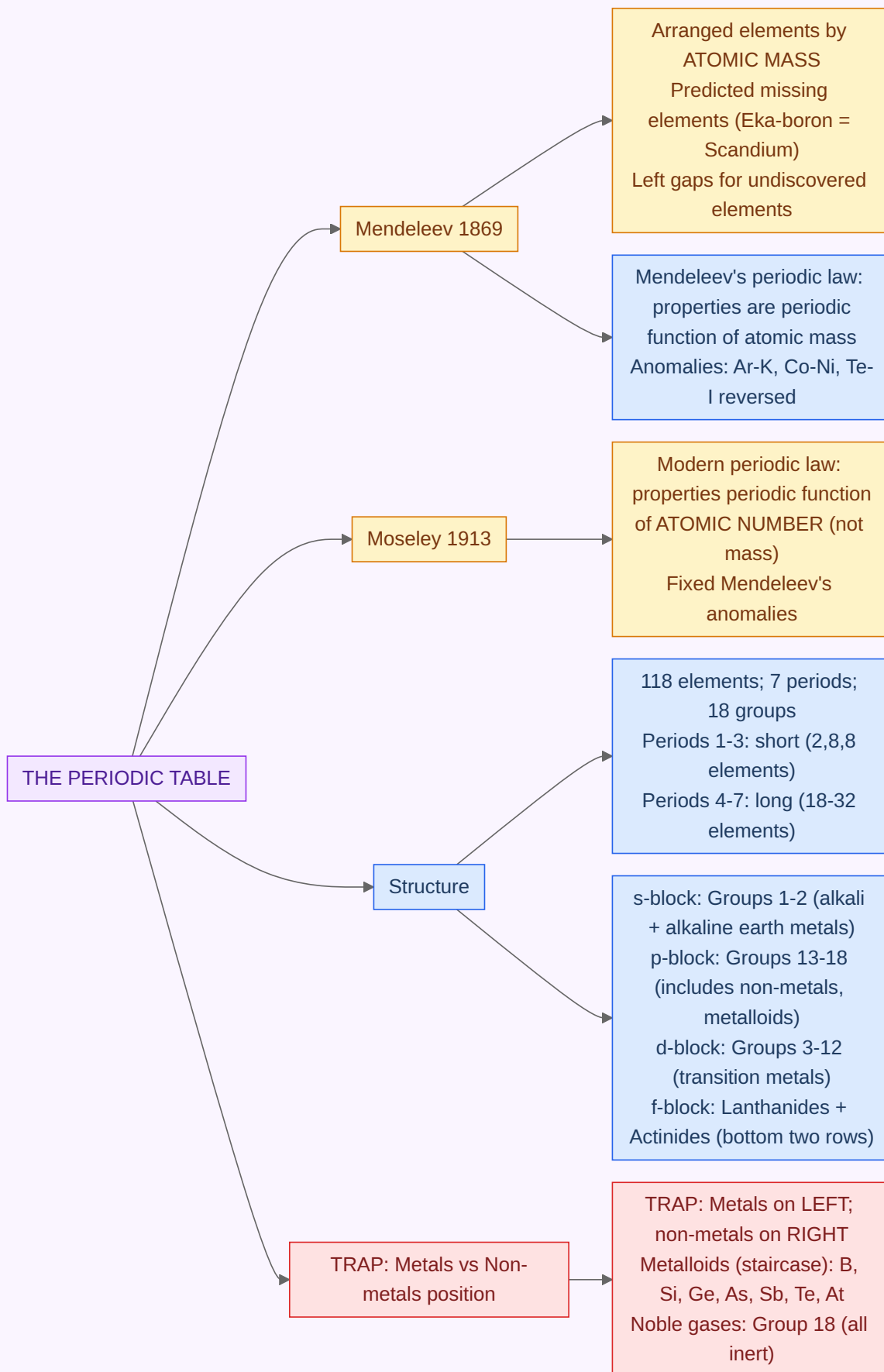
The most electronegative element is **Fluorine (F)**, **4.0 on Pauling scale**. The most reactive non-metal is Fluorine. The most reactive metal is **Caesium (Cs)** — sometimes Francium is given as the answer, but Cs is more commonly tested. The lightest metal is **Lithium (Li)**. The only liquid metal at room temperature is **Mercury (Hg)** — Gallium is also liquid near room temperature (melts at 29.8 °C) and appears as a trap option.

REMEMBER

Electronegativity decreasing order: **F > O > N > Cl > Br > I**. Memory: "**French Ocean Navy Cleans Burmese Islands.**"

PART C — CHEMICAL BONDING AND REACTIONS

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Chapter C1 — How Atoms Bond Together

Three main bond types — knowing which type forms tells you almost everything about the compound.

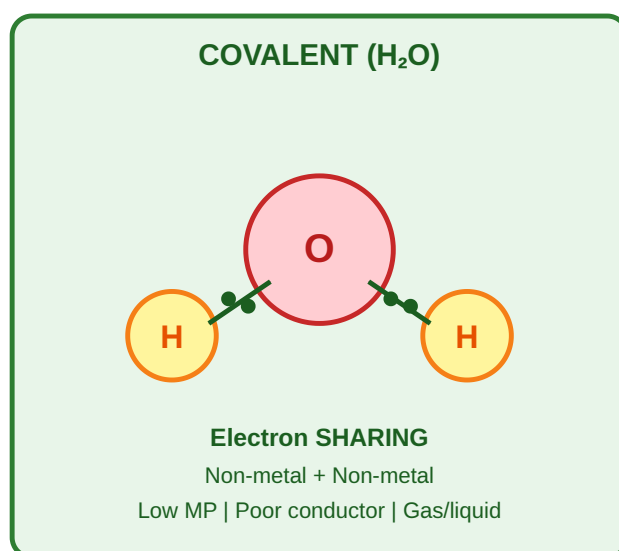
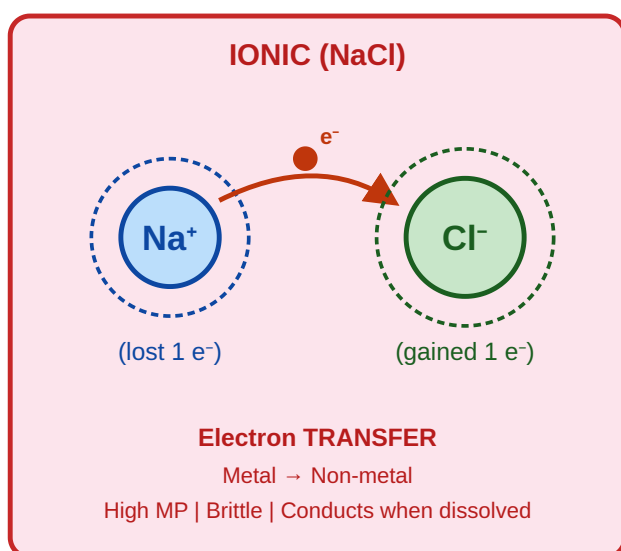
Ionic Bond — by transfer

- One atom **gives** electron(s), another **takes**.
- Result: oppositely charged ions attract.
- Typically forms between a **metal + non-metal**.
- **Examples** — NaCl (table salt), MgO, CaF₂.
- **Properties** — high MP, conducts when dissolved/molten, brittle.

Covalent Bond — by sharing

- Two non-metals **share** electron pairs.
- **Single** (1 pair), **double** (2 pairs), **triple** (3 pairs).
- **Examples** — H₂O, CH₄, O₂ (double), N₂ (triple).
- **Properties** — low MP, poor conductor, often gas or liquid.

Ionic vs Covalent Bonding



Metallic Bond — "sea of electrons"

- Valence electrons form a **delocalised sea** that flows freely around positive metal cores.
- This explains why metals: **conduct electricity + heat, are malleable, and are lustrous**.

Dative (Coordinate) Bond

- Both electrons in the shared pair come from the **same atom**.
- **Examples** — NH₄⁺ (ammonium), H₃O⁺ (hydronium).

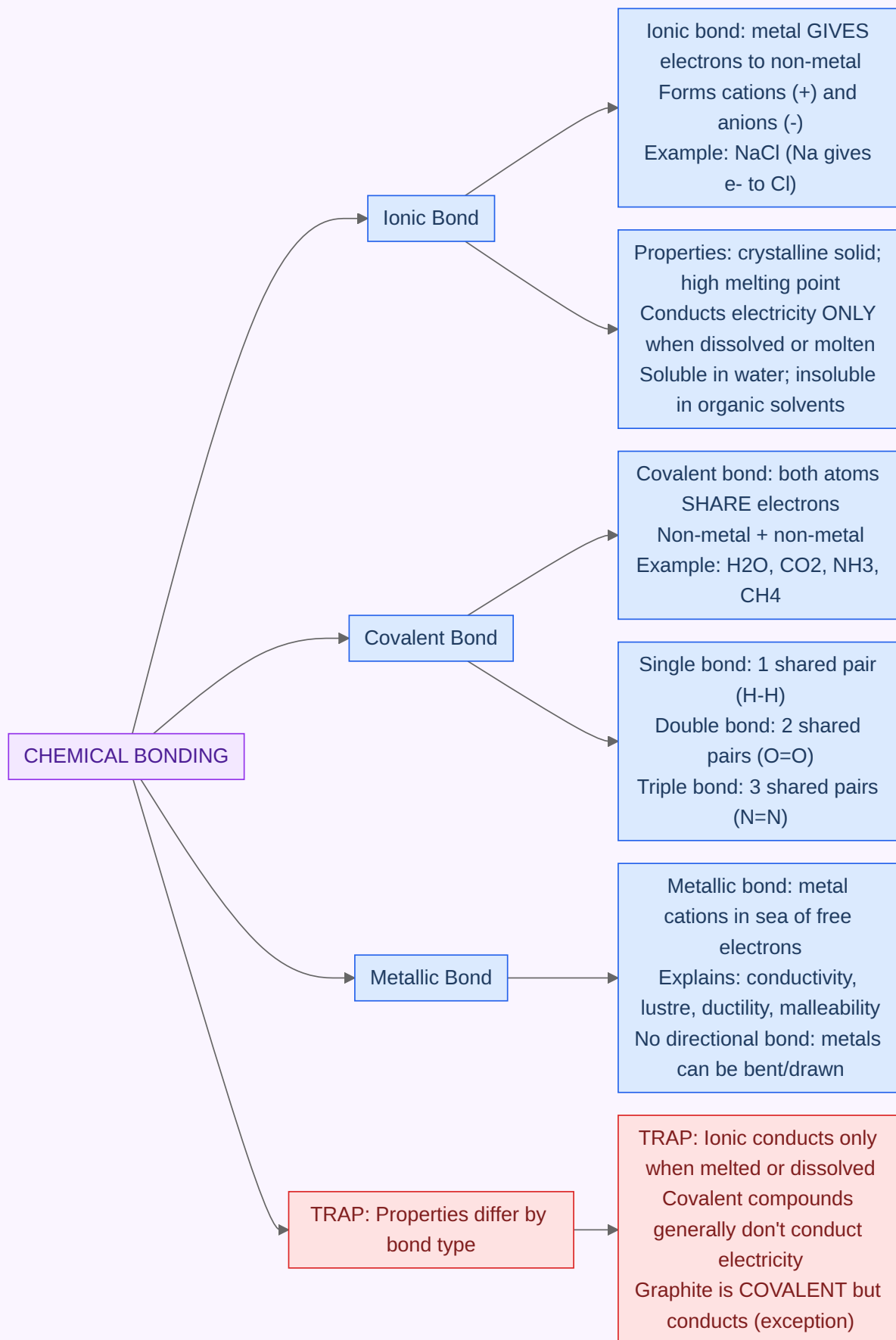
Hydrogen Bond

- A weak intermolecular attraction.
- Forms when H is bonded to a very electronegative atom — **F, O, or N only**.
- Strength: ~1/10 of a covalent bond, but enough to keep water liquid at room temp.

KEY FACT

Hydrogen bonding explains why water is liquid at room temperature (small molecule but high boiling point), why ice floats on water (hydrogen bonds form an open hexagonal lattice in ice → lower density than liquid water), and how DNA bases pair (A-T with 2 H-bonds, G-C with 3 H-bonds).

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Chapter C2 — Types of Chemical Reactions

Every exam reaction question maps to one of these categories. Know them by name and example.

Type	Pattern	Example
Combination	$A + B \rightarrow AB$	$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$
Decomposition	$AB \rightarrow A + B$	$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
Displacement	$A + BC \rightarrow AC + B$	$\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
Double displacement	$AB + CD \rightarrow AD + CB$	$\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl}\downarrow + \text{NaNO}_3$
Redox	electron transfer	$\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$ (rusting)
Combustion	$\text{fuel} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{heat}$	$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
Exothermic	releases heat	neutralisation, combustion
Endothermic	absorbs heat	photosynthesis, thermal decomposition

REMEMBER

Oxidation and reduction — **OIL RIG**: Oxidation Is Loss (of electrons), Reduction Is Gain (of electrons). In the reaction $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$, Zn loses electrons (oxidised), Cu^{2+} gains them (reduced). Zinc is the reducing agent; Cu^{2+} is the oxidising agent.

Analogy: Think of electrons like money. Oxidation = losing your money (going poor = losing electrons). Reduction = gaining money (getting rich = gaining electrons). The person who steals your money (oxidising agent) gets reduced (gains electrons); you get oxidised (lose electrons).

EXAM ALERT

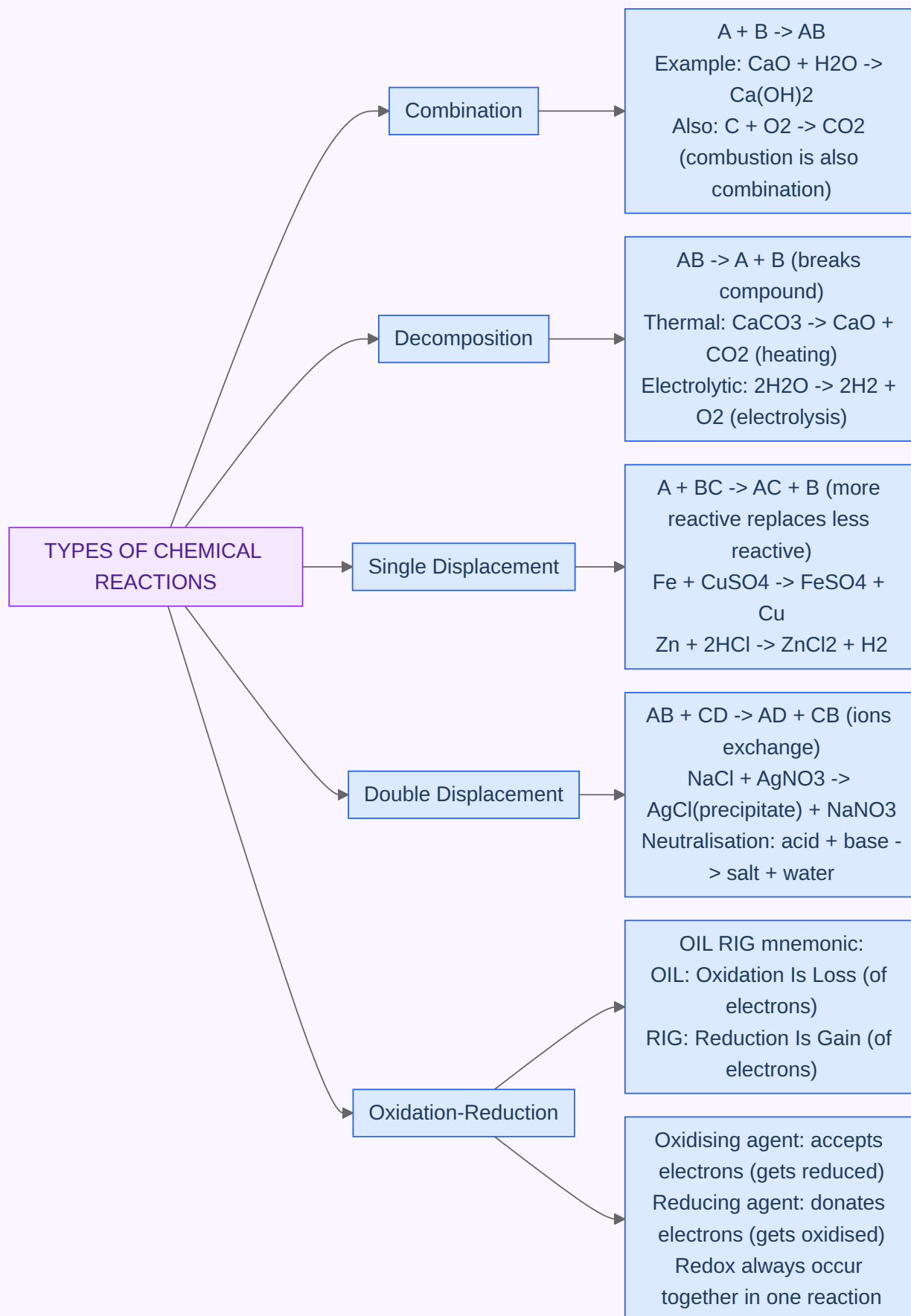
⚠ TRAP: "The reducing agent is oxidised" and "the oxidising agent is reduced" — this confuses almost every student. The reducing agent **GIVES** electrons → it gets oxidised. The oxidising agent **TAKES** electrons → it gets reduced. In rusting: iron is the reducing agent ($\text{Fe} \rightarrow \text{Fe}^{3+} + 3\text{e}^-$), oxygen is the oxidising agent (gains electrons).

COMMON TRAP

Students confuse "oxidation" with "gaining oxygen" — that is only the old, incomplete definition. The modern definition is about electron loss/gain. Rusting is a redox reaction (iron loses electrons), not just "combining with oxygen."

PART D — ACIDS, BASES, SALTS

⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW

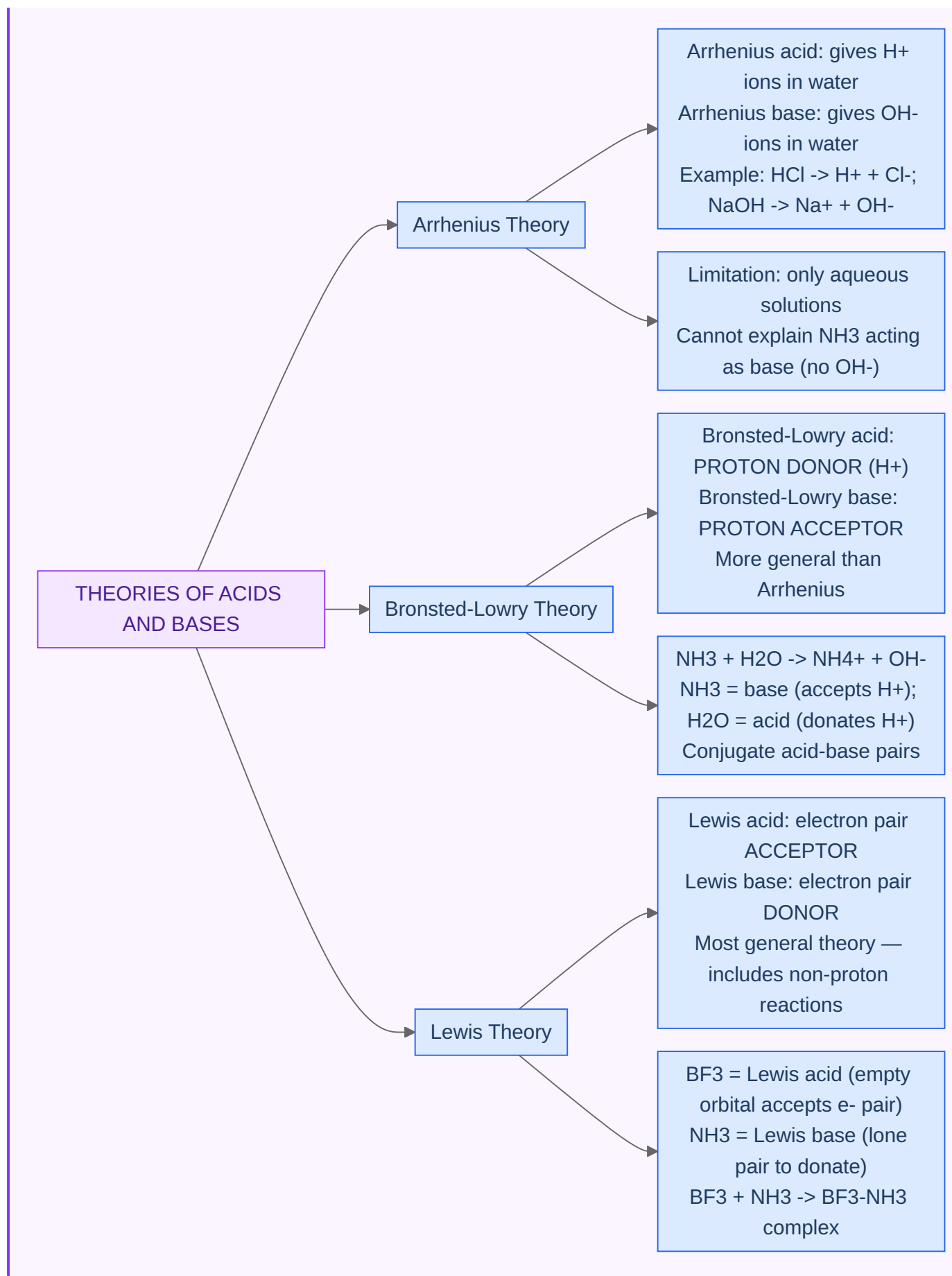


Chapter D1 — Three Theories of Acids and Bases

Theory	Acid	Base	Limitation
Arrhenius (1884)	Gives H^+ in water	Gives OH^- in water	Only for water solutions
Brønsted-Lowry (1923)	H^+ donor	H^+ acceptor	Needs H^+ transfer
Lewis (1923)	Electron-pair acceptor	Electron-pair donor	Most general; covers all

Lewis theory is the broadest. AlCl_3 is a Lewis acid (accepts electrons) even though it has no H^+ to give.

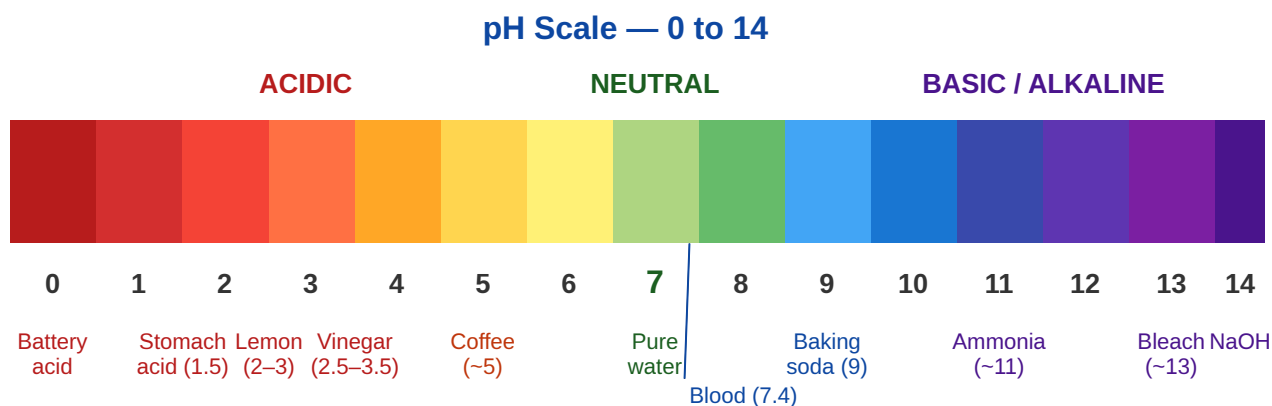
⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter D2 — The pH Scale

The idea

- $\text{pH} = -\log[\text{H}^+]$ (negative log of hydrogen-ion concentration).
- A **0-to-14 ruler** where 0 is maximally acidic and 14 is maximally basic.
- Each step = **10× change** in H^+ concentration.
- pH 7 = neutral; below = acidic; above = basic (alkaline).



Exam trap: Blood is NOT neutral!

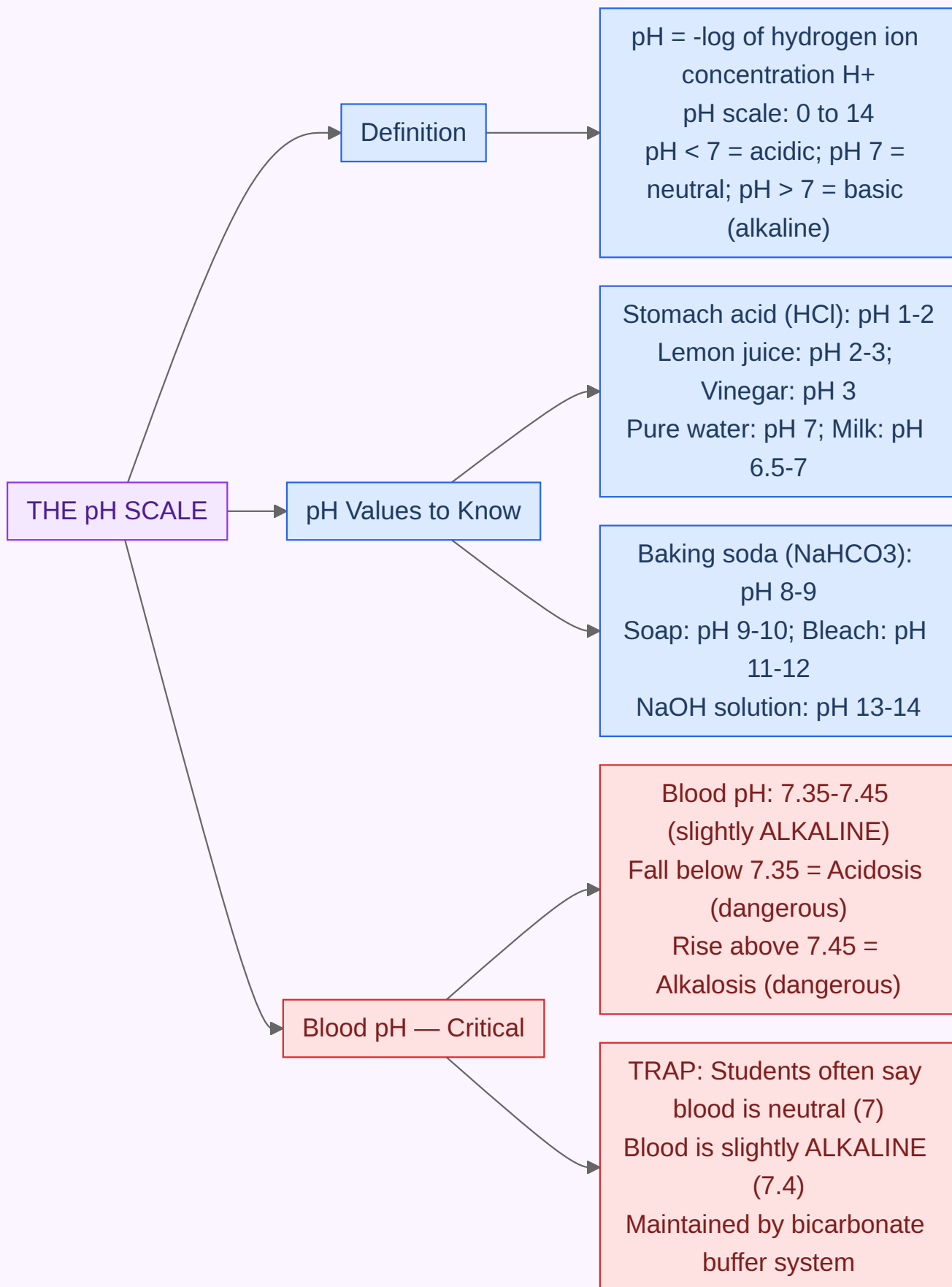
Blood = 7.35–7.45 (slightly alkaline). Below 7.35 = acidosis. Above 7.45 = alkalosis.
Bicarbonate buffer keeps blood pH steady.

Common pH values to memorise:

Substance	Approximate pH
Battery acid (H_2SO_4)	0
Stomach acid (gastric juice)	1.5–2
Lemon juice	2–3
Vinegar	2.5–3.5
Soft drinks (cola)	3
Tomato juice	4
Black coffee	5
Milk	6.5–6.8
Pure water	7
Blood	7.35–7.45
Sea water	8
Baking soda solution	9
Ammonia (household)	11
Bleach	13
Caustic soda (NaOH)	14

EXAM ALERT

Blood pH is 7.35–7.45 — slightly alkaline. A drop below 7.35 causes acidosis; above 7.45 causes alkalosis, both life-threatening. Gastric juice at pH 1.5–2 is almost as acidic as battery acid — yet your stomach lining survives because of a mucus coating.



Chapter D3 — Indicators (Learn the Colour Changes)

Indicator	In Acid	In Neutral	In Base
Litmus	Red	Purple	Blue
Phenolphthalein	Colourless	Colourless	Pink
Methyl orange	Red	Orange	Yellow
Methyl red	Red	Orange	Yellow
Bromothymol blue	Yellow	Green	Blue
Universal indicator	Red→Orange	Green	Blue→Violet
Turmeric (natural)	Yellow	Yellow	Red - brown

REMEMBER

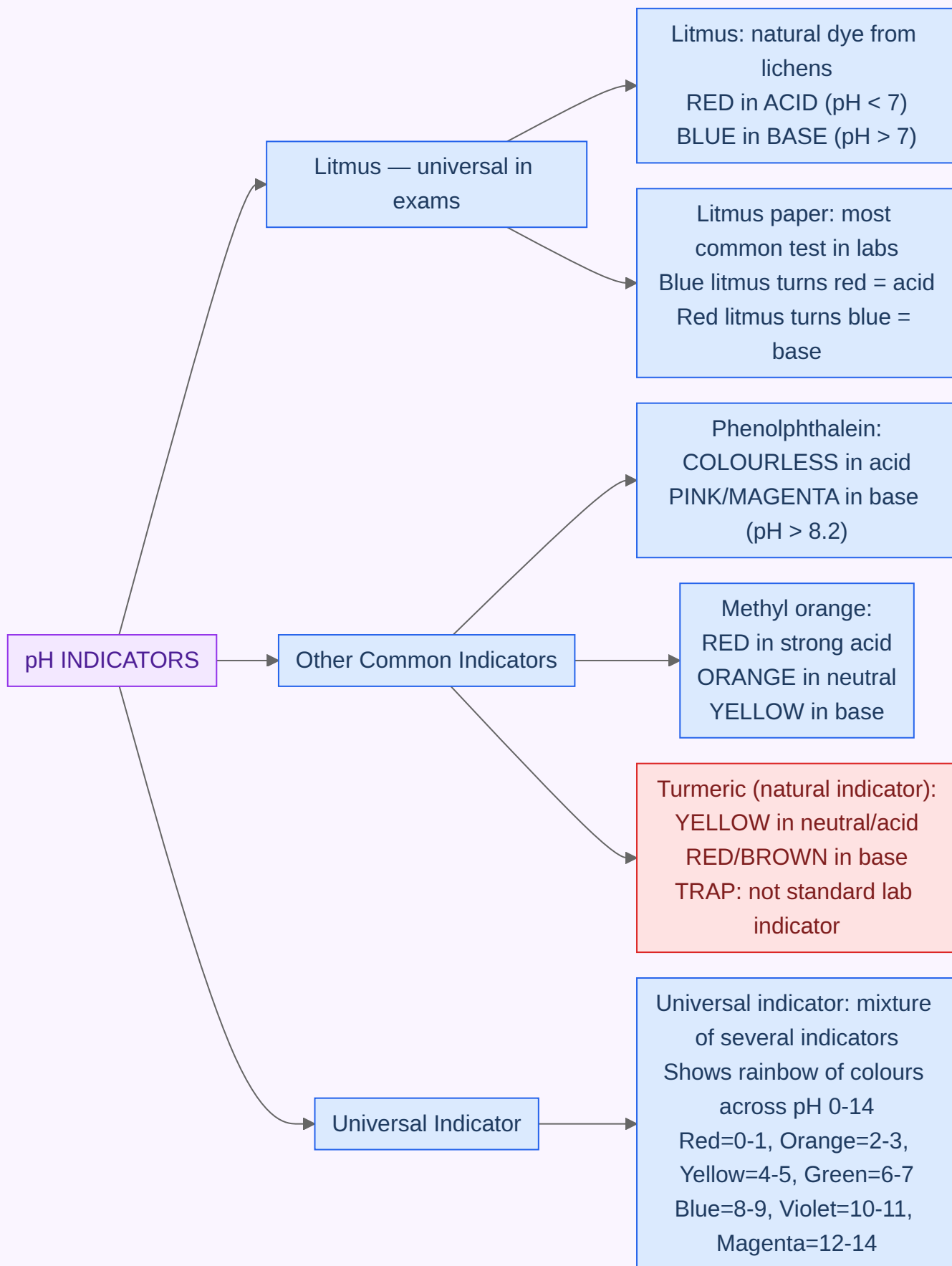
Litmus: **A-Red, B-Blue** — Acids turn litmus Red, Bases turn it Blue. Phenolphthalein: **colourless in acid, PiNk in base** — the P is for Pink in base.

Quick shortcut for indicators: Litmus and Methyl Orange both turn **Red** in acid. Phenolphthalein turns **Pink** in base (colourless in acid — the only one that "disappears" in acid). Universal indicator is the rainbow — Red=acid, Green=neutral, Violet=strong base.

EXAM ALERT

⚠ TRAP: Phenolphthalein is colourless in acid AND in neutral. It only turns pink at $\text{pH} \geq 8.2$. Students often write "yellow in acid" (that's methyl orange) or "red in acid" (that's litmus). Phenolphthalein in acid = **colourless**. Also: turmeric is a natural indicator but NOT used in standard titrations — it changes too slowly.

CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter D4 — Common Acids and Their Uses

Acid	Formula	Common name / use
Hydrochloric acid	HCl	Spirit of salt; gastric juice; metal cleaning; descaler
Sulphuric acid	H ₂ SO ₄	Oil of vitriol; "king of chemicals"; car batteries; fertilisers
Nitric acid	HNO ₃	Aqua fortis; explosives; fertilisers
Acetic acid	CH ₃ COOH	Vinegar; food preservative
Citric acid	C ₆ H ₈ O ₇	Lemons, oranges; sour taste
Tartaric acid	C ₄ H ₆ O ₆	Tamarind; baking powder ingredient
Oxalic acid	H ₂ C ₂ O ₄	Tomatoes, spinach; stain remover
Lactic acid	CH ₃ CH(OH)COOH	Curd; sour milk
Formic acid	HCOOH	Ant sting; bee venom

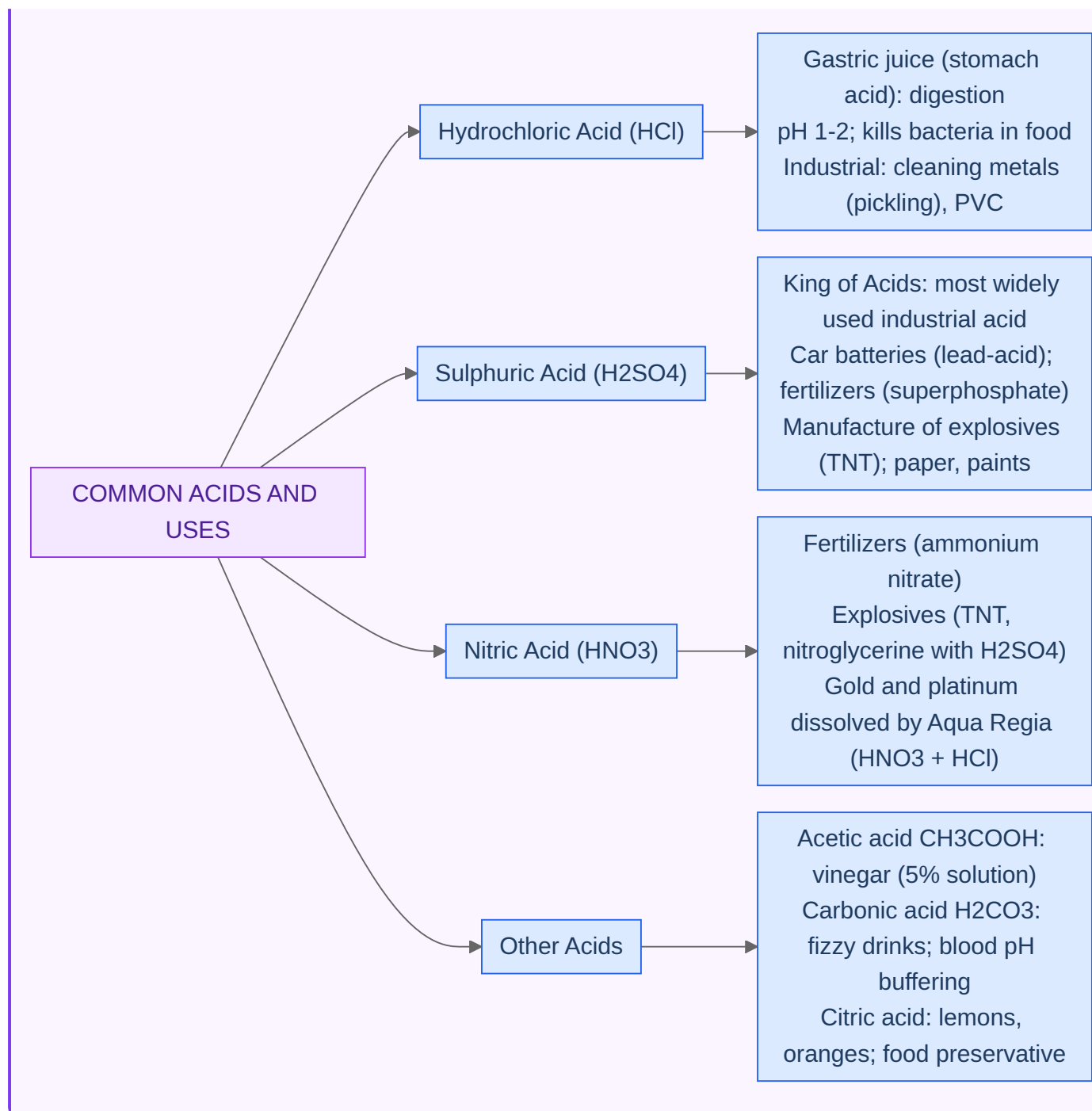
KEY FACT

Aqua regia = 3 parts HCl + 1 part HNO₃ (by volume). It dissolves gold and platinum — the only common mixture that can. This question appears in almost every competitive exam's chemistry section.

REMEMBER

Acid nicknames to memorise cold: HCl = "Spirit of Salt" | H₂SO₄ = "Oil of Vitriol" / "King of Chemicals" | HNO₃ = "Aqua Fortis" (strong water) | Aqua Regia = "Royal Water" (dissolves the kings of metals: gold and platinum). Memory hook: **"The King dissolved in Royal Water"** = Gold dissolved in Aqua Regia.

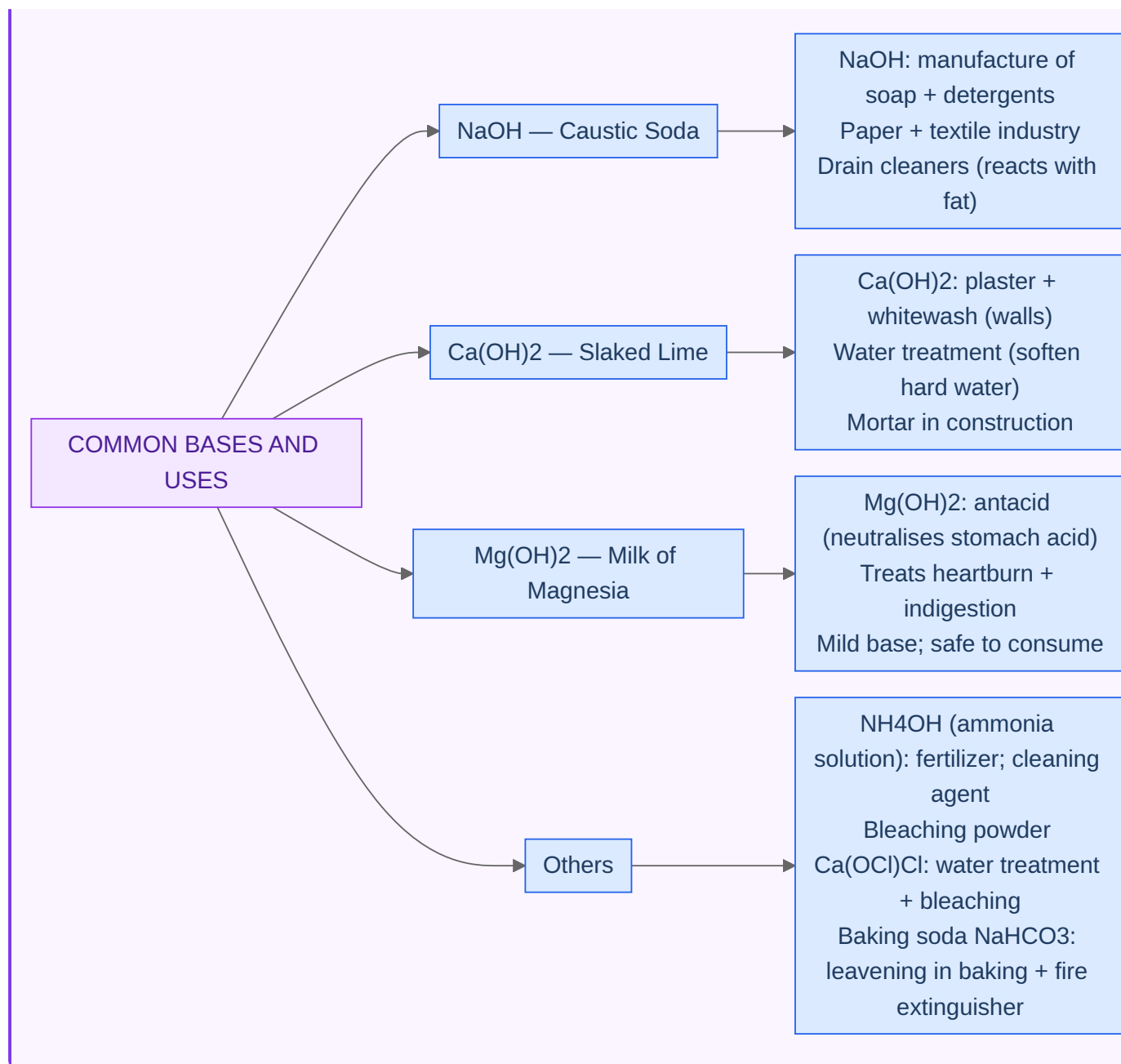
CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter D5 — Common Bases and Their Uses

Base	Formula	Common name / use
Sodium hydroxide	NaOH	Caustic soda; soap-making, paper
Potassium hydroxide	KOH	Caustic potash; soft soap, batteries
Calcium hydroxide	Ca(OH) ₂	Slaked lime; whitewash, mortar
Magnesium hydroxide	Mg(OH) ₂	Milk of magnesia; antacid
Ammonium hydroxide	NH ₄ OH	Household cleaner

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter D6 — Salts and Their pH

Salt	Acid + Base used	pH
Sodium chloride (NaCl)	Strong + Strong	Neutral (7)
Ammonium chloride (NH ₄ Cl)	Weak base + Strong acid	Acidic (<7)
Sodium carbonate (Na ₂ CO ₃)	Strong base + Weak acid	Basic (>7)
Sodium bicarbonate (NaHCO ₃)	Strong base + Weak acid	Slightly basic (8.3)

EXAM ALERT

Baking soda is NaHCO₃ (sodium bicarbonate). It is mildly basic (pH ~8.3). When you eat something acidic, taking baking soda neutralises the acidity in your stomach. Also used in fire extinguishers — it decomposes on heating to release CO₂ which smothers fires.

REMEMBER

Common sodium salts — 4 to memorise cold: - **NaCl** = Table salt (common salt, neutral pH 7) - **NaHCO₃** = Baking soda (Sodium biCARBonate — the HCO₃ part tells you it's bicarbonate) - **Na₂CO₃** = Washing soda (the 2 carbonate = stronger/heavier = washing) - **NaOH** = Caustic soda (Hydroxide — the only "OH" = strong base, caustic)

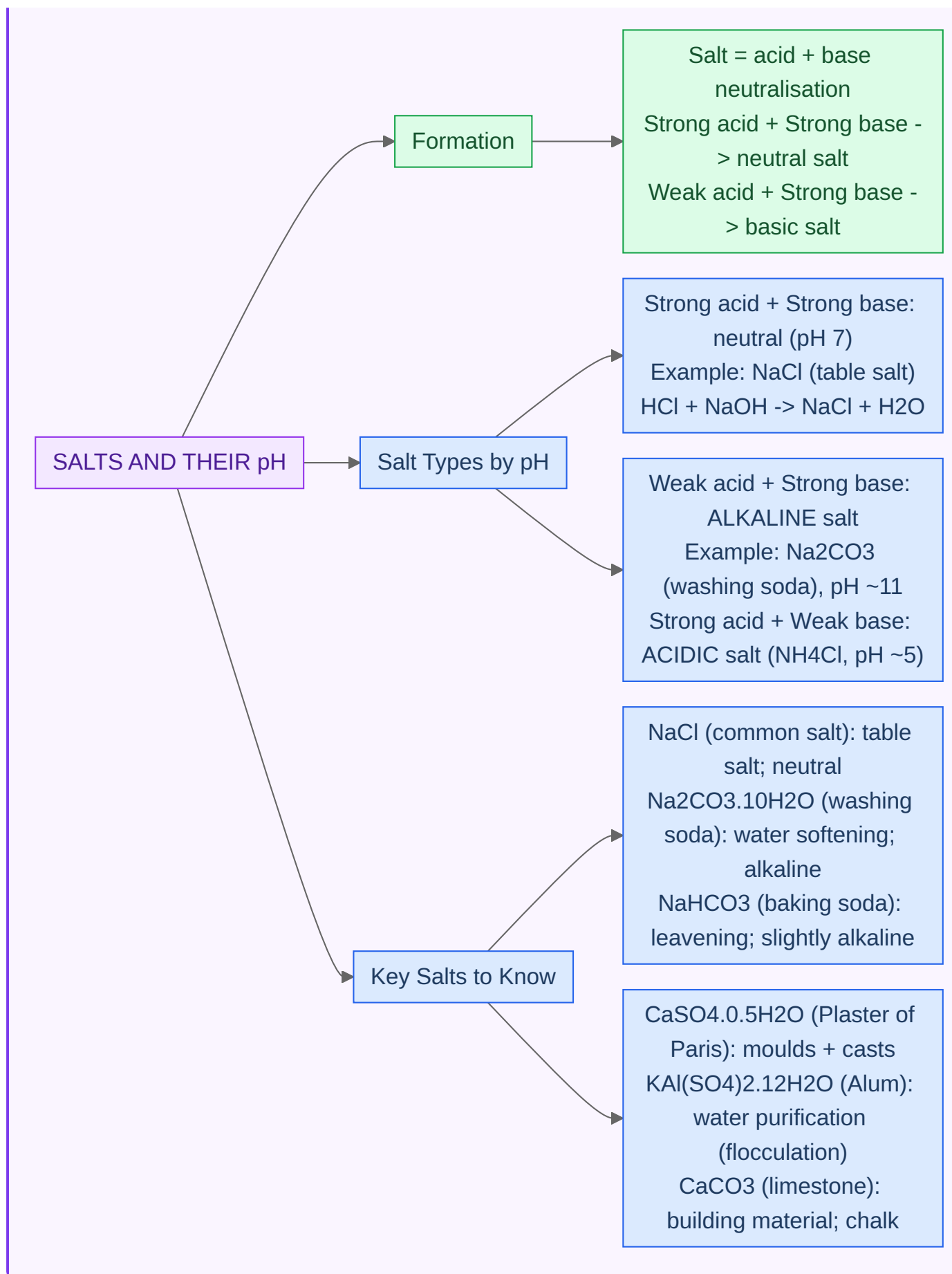
Quick rule: **biCARBonate = baking; CARBonate = washing**. One carbon less (HCO₃ vs CO₃) makes it milder (baking vs washing).

EXAM ALERT

⚠ TRAP — Baking soda vs Baking powder: Baking soda = pure NaHCO₃. Baking powder = NaHCO₃ + tartaric acid + starch. Baking powder has its own acid, so it doesn't need an acidic ingredient in the recipe. Baking soda needs an acid (lemon juice, buttermilk) to react. Examiners ask: "What is baking powder made of?" — not just baking soda.

PART E — METALS AND THE ACTIVITY SERIES

🕒 CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



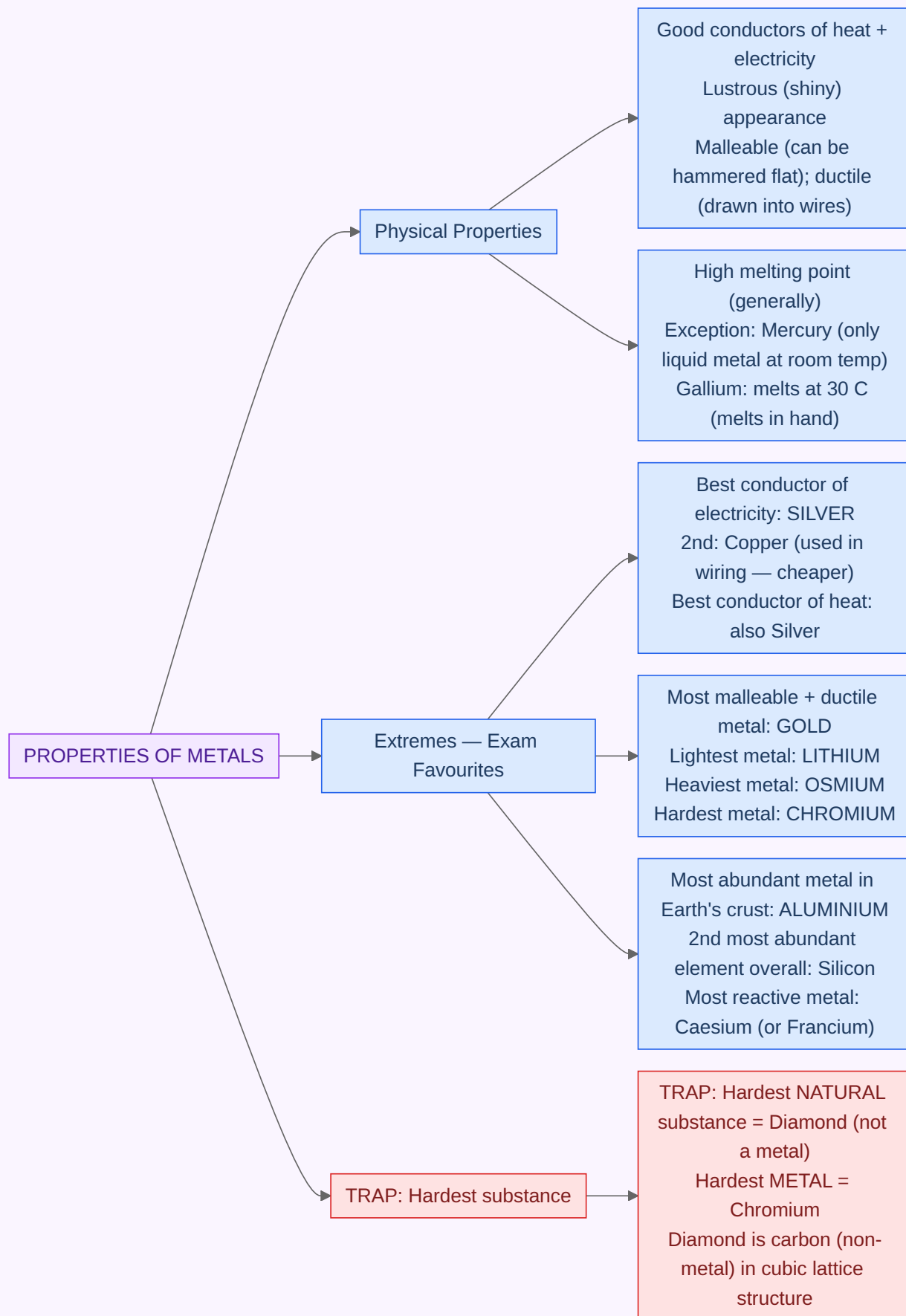
Chapter E1 — Properties of Metals

General properties: Lustrous, malleable, ductile, good conductors of heat and electricity, typically solid at room temperature with high melting points.

Important exceptions — examiners love these:

- **Mercury (Hg)** — only metal liquid at room temperature. (Gallium also melts near room temperature at 29.8 °C — a common trap.)
- **Sodium (Na) and Potassium (K)** — metals but so soft you can cut them with a knife; stored in kerosene oil to prevent reaction with air/water.
- **Gold (Au)** — most malleable and ductile metal. 1 gram of gold can be drawn into 2.4 km of wire.
- **Silver (Ag)** — best conductor of electricity. We use copper in wiring only because silver is too expensive.
- **Tungsten (W)** — highest melting point among all metals (3422 °C). Used in light bulb filaments.
- **Lithium (Li)** — lightest metal (density 0.534 g/cm³).
- **Osmium (Os)** — densest naturally occurring metal.

CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter E2 — Activity Series (Reactivity Series)

One of the most frequently tested topics in competitive chemistry. Memorise the order — questions just rephrase it.

Most reactive → Least reactive:

$K > Na > Ca > Mg > Al > Zn > Fe > Sn > Pb > [H] > Cu > Hg > Ag > Au > Pt$

KEY FACT

Quick Recall — Activity Series: The series tells you three things instantly: (1) Who displaces whom in a reaction. (2) Which metals react with water / acids. (3) Which metals need electrolysis to extract. Metals above hydrogen react with dilute acids and liberate H_2 . Metals below hydrogen (Cu, Hg, Ag, Au, Pt) don't react with dilute acids.

REMEMBER

"Please Stop Calling Me A Zebra, I Stay Planted Here, Cause He's Still At Practice"

P = Potassium (K), S = Sodium (Na), C = Calcium (Ca), M = Magnesium (Mg), A = Aluminium (Al), Z = Zinc (Zn), I = Iron (Fe), S = Tin (Sn), P = Lead (Pb), H = Hydrogen, C = Copper (Cu), H = Mercury (Hg), S = Silver (Ag), A = Gold (Au), P = Platinum (Pt)

What the Series Tells You

- Metals above hydrogen in the series can displace H from acids. Metals below hydrogen (Cu, Hg, Ag, Au, Pt) cannot — they do not react with dilute acids.
- A metal higher in the series displaces a lower one from its salt solution. This is why $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ (zinc displaces copper because Zn is higher than Cu).
- K, Na react with cold water violently. Ca and Mg react with hot water. Fe reacts with steam.

COMMON TRAP

"Most reactive metal" — the answer examiners want is Caesium (Cs) or Potassium (K) depending on how the question is framed. Francium (Fr) is also in Group 1 and is technically most reactive, but it is synthetic and radioactive, so practical exams usually mean Cs or K.

CHAPTER SUMMARY — EVERYTHING IN ONE VIEW

ACTIVITY SERIES OF METALS

Order of Reactivity

$K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au > Pt$
Decreasing reactivity left to right

Above H: displace H_2 from dilute acid
Above Ca: even displace H_2 from cold water
Below H (Cu, Hg, Ag, Au): do NOT react with dilute acids

Rules for Reactions

More reactive metal displaces less reactive from salt solution
 $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (Fe more reactive than Cu)
 $Cu + FeSO_4 \rightarrow$ NO REACTION (Cu less reactive than Fe)

Gold + Platinum: most stable; no tarnishing
Noble metals; very hard to react
Used in jewellery and electrical contacts

TRAP: Thermite reaction

Thermite reaction: $Al + Fe_2O_3 \rightarrow Al_2O_3 + Fe + \text{heat}$
Al is more reactive than Fe (displaces it)
Used in WELDING of railway tracks (exothermic)

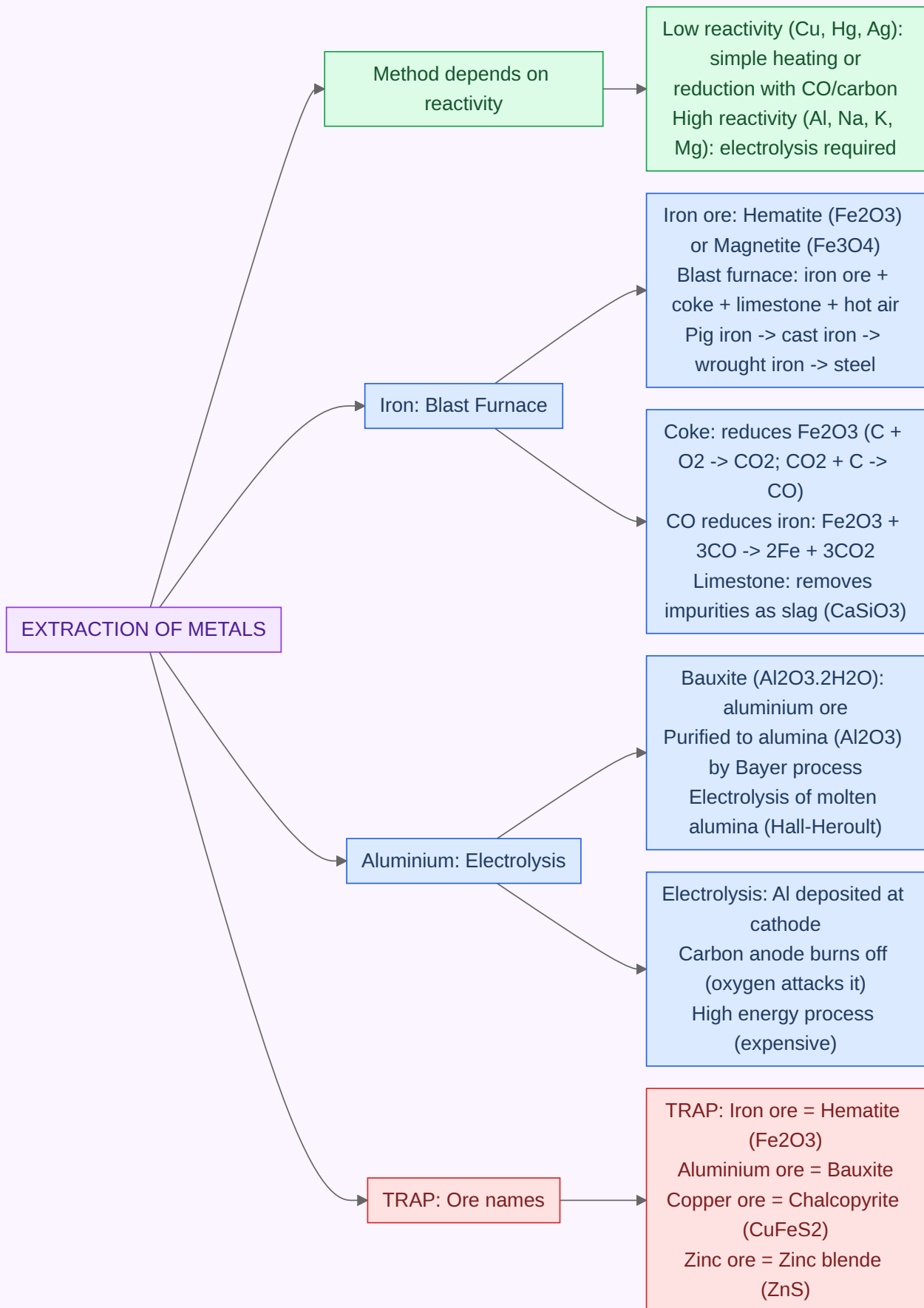
Chapter E3 — Extraction of Metals

Metal	Ore (common)	Extraction method
Aluminium	Bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$)	Bayer process (concentration) + Hall-Héroult (electrolysis in molten cryolite)
Iron	Haematite (Fe_2O_3)	Blast furnace with coke
Copper	Copper pyrites (CuFeS_2)	Roasting + smelting + Bessemer process
Zinc	Zinc blende (ZnS)	Roasting + carbon reduction
Lead	Galena (PbS)	Roasting + reduction
Mercury	Cinnabar (HgS)	Direct heating — Hg vapour condenses
Sodium	Rock salt (NaCl)	Down's process (electrolysis of molten NaCl)
Silver	Argentite (Ag_2S)	Cyanide leaching
Gold	Native gold	Cyanidation with NaCN solution
Tin	Cassiterite (SnO_2)	Carbon reduction

KEY FACT

Hall-Héroult process (1886): Electrolysis of alumina (Al_2O_3) dissolved in molten cryolite (Na_3AlF_6). This is how all commercial aluminium is made. Cryolite is the solvent that lowers the melting point from 2050°C to around 950°C , making the process practical.

CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter E4 — Corrosion and Prevention

- **Rusting (iron):** $\text{Fe} + \text{O}_2 + \text{H}_2\text{O} \rightarrow \text{hydrated iron oxide (Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O)}$ — the red-brown flaky coating. Both oxygen AND water are needed; iron does not rust in dry air or pure water alone.
- **Tarnishing (silver):** $\text{Ag} + \text{H}_2\text{S} \rightarrow \text{Ag}_2\text{S}$ (black coating).
- **Patina (copper/bronze):** Green coating on copper statues = basic copper carbonate.

Prevention methods: - Galvanising — coating iron with zinc. Even if the Zn coat is scratched, Zn corrodes preferentially (sacrificial anode). - Tinning — coating with tin (food cans). If scratched, iron corrodes faster. - Electroplating — chrome, nickel. - Alloying — stainless steel (Cr + Ni) is naturally corrosion-resistant. - Painting, oiling, varnishing.

EXAM ALERT

Galvanising uses zinc as a protective coat. Zinc is above iron in the activity series, so zinc acts as a sacrificial anode — it oxidises preferentially, protecting the iron. This is cathodic protection. Tinning uses tin (below iron in the series), so if the tin coat is scratched, iron corrodes faster.

REMEMBER

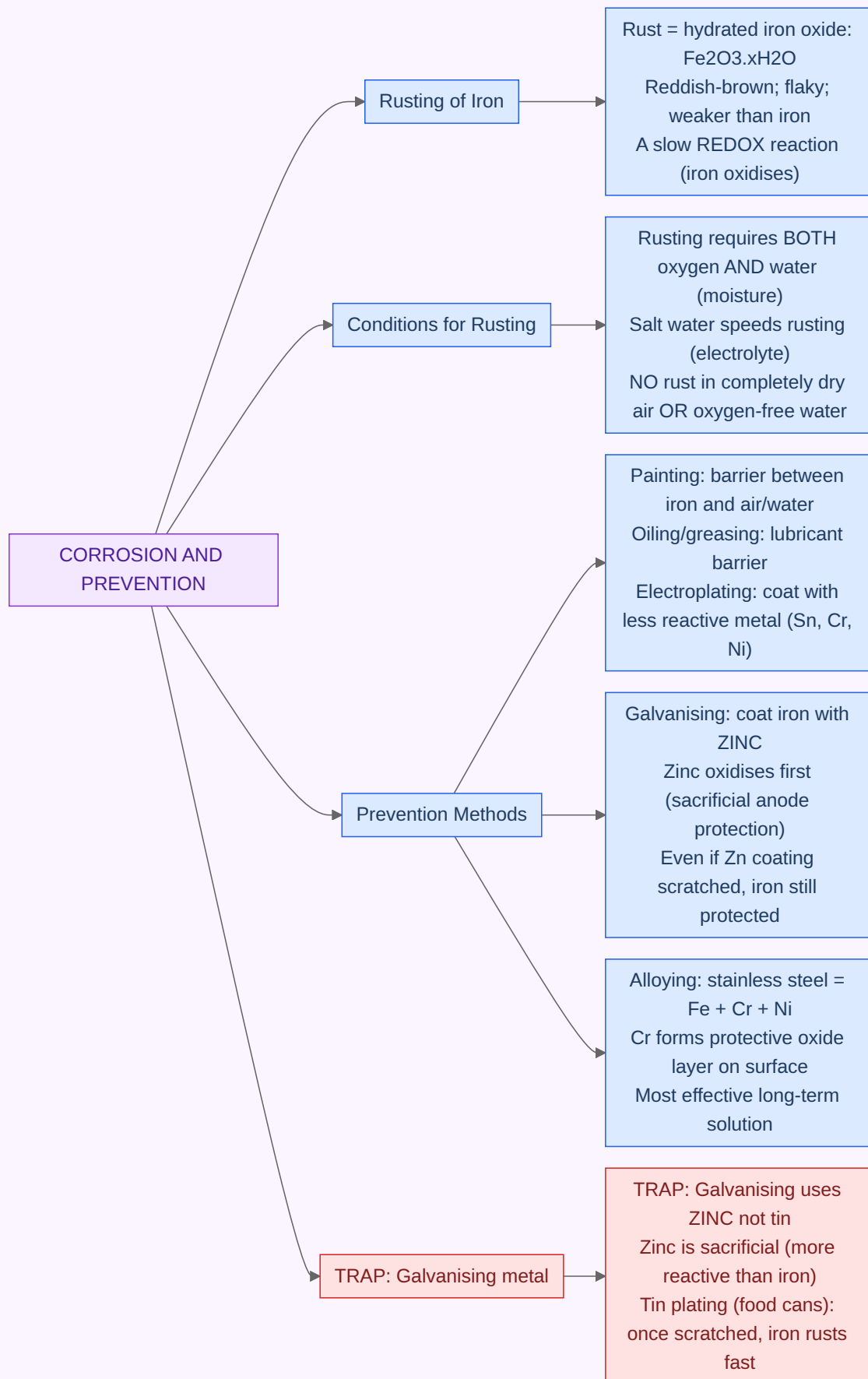
"Galvanising = Zinc Protects Iron; Tinning = Tin Betrays Iron." Galvanising: Zinc is more reactive → it corrodes first → iron stays safe (even if zinc is scratched). Tinning: Tin is less reactive → once scratched, iron is exposed and corrodes faster. Remember: **Galvanise = Guardian** (zinc guards iron). **Tinning = Traitor** (tin betrays iron when damaged).

EXAM ALERT

⚠ TRAP: "Both galvanising and tinning protect iron." True as long as the coat is intact — but the MECHANISM is opposite. Galvanising: zinc corrodes sacrificially even if coat is scratched (safe). Tinning: tin coat must be unbroken — once scratched, iron rusts faster (unsafe). Examiners test this contrast directly.

PART F — ORGANIC CHEMISTRY

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter F1 — Why Carbon Is Special

Carbon forms 4 bonds (tetravalent), chains indefinitely with itself (catenation), and forms stable double and triple bonds. Result: over 10 million known organic compounds versus about 500,000 for all other elements combined. No other element comes close.

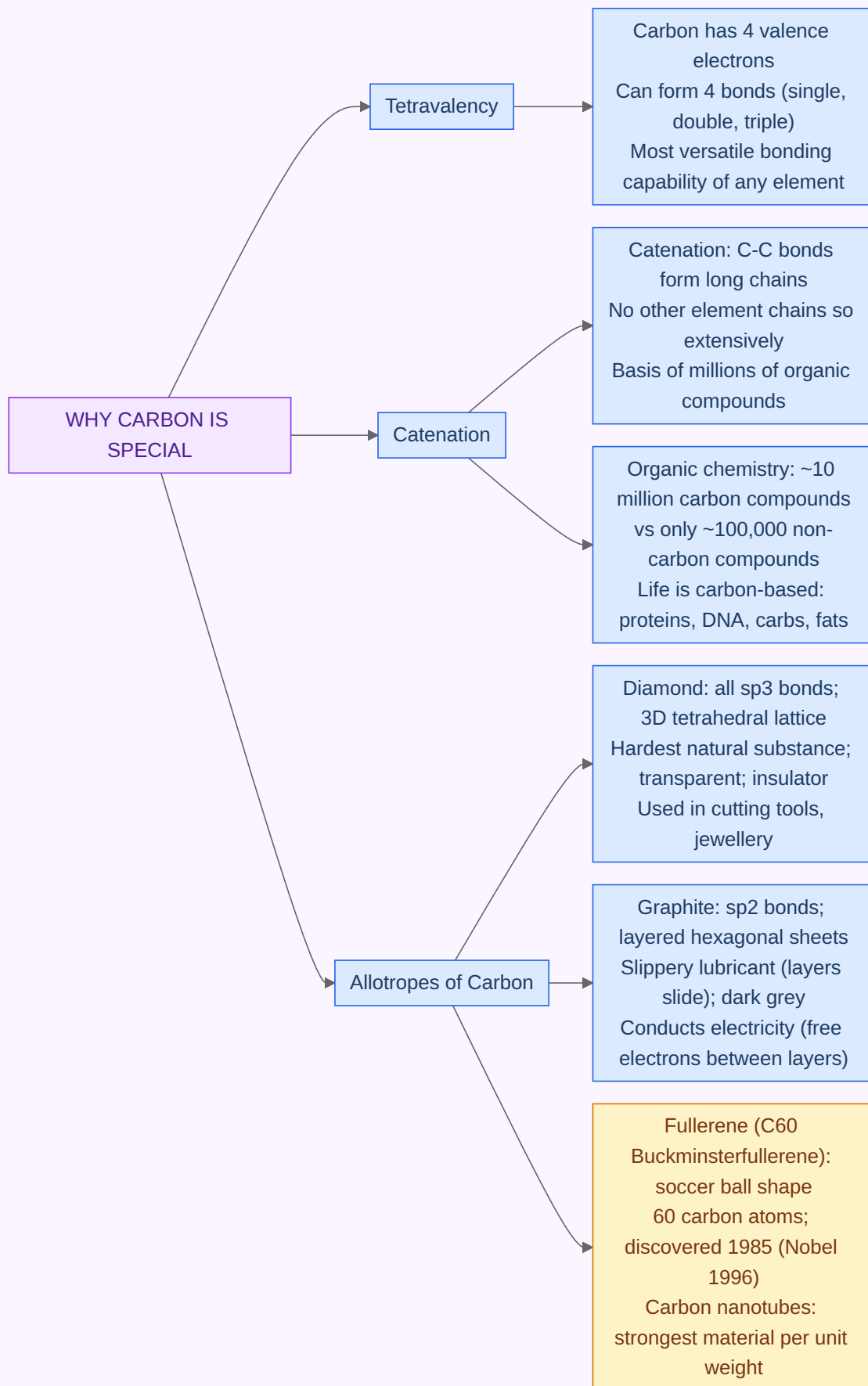
KEY FACT

Quick Recall — Carbon allotropes: Diamond (hardest, insulator, 3D tetrahedral lattice) vs Graphite (soft, conductor, layered hexagonal sheets). Both are pure carbon — just different arrangements. **"Diamond is for ever"** because its 3D bonds are unbreakable. Graphite layers slide (pencil writing) because only weak van der Waals forces hold sheets together.

EXAM ALERT

⚠ TRAP: "Graphite is a non-metal that conducts electricity" — this is the famous exception. Most non-metals are insulators. Graphite conducts because it has delocalised electrons between its hexagonal layers. Diamond (also pure carbon) is an insulator. The examiner knows students forget this and sets it regularly.

📄 CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter F2 — Hydrocarbons — The Three Series

Alkanes — C_nH_{2n+2} (single bonds, saturated)

Methane (CH_4), Ethane (C_2H_6), Propane (C_3H_8), Butane (C_4H_{10}), Pentane (C_5H_{12}), Hexane (C_6H_{14}), Heptane (C_7H_{16}), Octane (C_8H_{18}), Nonane (C_9H_{20}), Decane ($C_{10}H_{22}$).

REMEMBER

First 10 alkanes: "**Me Eats Peanut Butter, Pretty Happily; Hears Only Nice Decrees**" — Methane, Ethane, Propane, Butane, Pentane, Hexane, Heptane, Octane, Nonane, Decane.

Alkenes — C_nH_{2n} (one double bond, unsaturated)

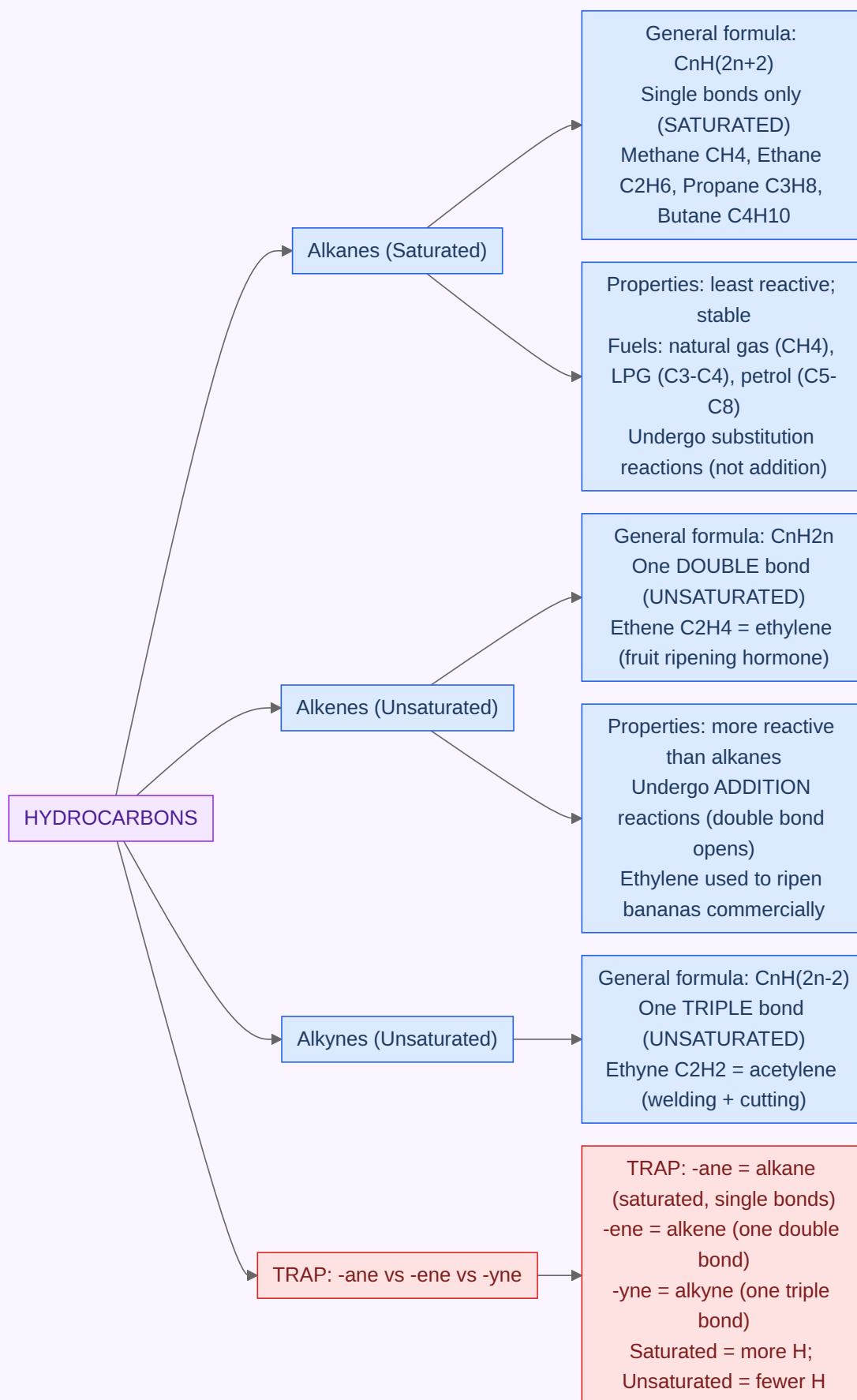
Ethene (C_2H_4), Propene (C_3H_6), Butene (C_4H_8)...

Ethene (ethylene) is the plant hormone that triggers fruit ripening. This is why a ripe banana in a fruit bowl makes other fruits ripen faster.

Alkynes — C_nH_{2n-2} (one triple bond, unsaturated)

Ethyne (acetylene, C_2H_2) is the most important. It burns at over $3000\text{ }^{\circ}C$ with oxygen — used in welding (oxyacetylene torch).

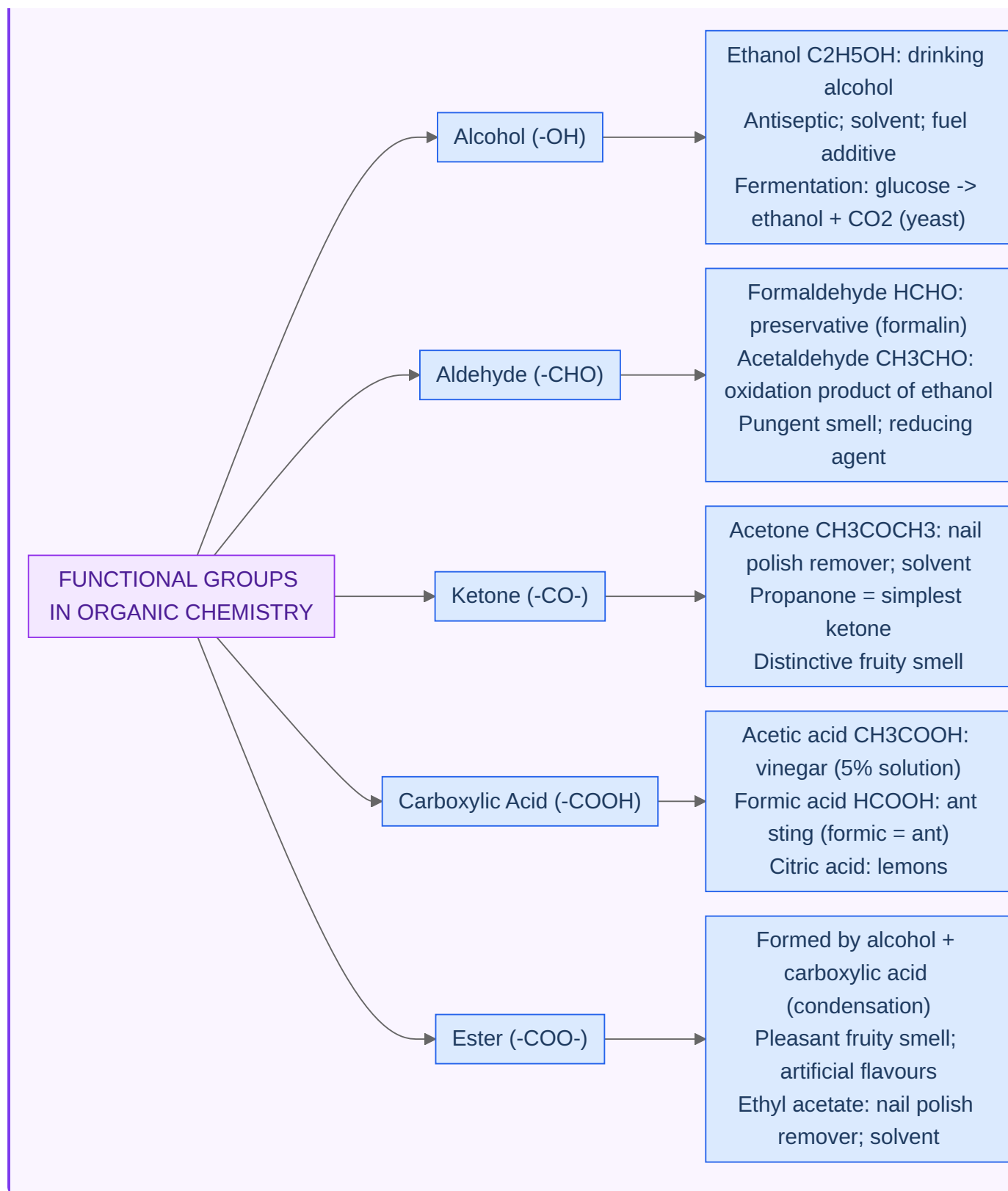
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Chapter F3 — Functional Groups

Group	Formula	Class	Key Example
Hydroxyl	-OH	Alcohol	Ethanol (C ₂ H ₅ OH) — drinking alcohol
Aldehyde	-CHO	Aldehyde	Formaldehyde (HCHO) — preservative
Ketone	>C=O	Ketone	Acetone (CH ₃ COCH ₃) — nail polish remover
Carboxyl	-COOH	Carboxylic acid	Acetic acid (CH ₃ COOH) — vinegar
Ester	-COO-	Ester	Ethyl acetate — nail polish remover solvent
Amine	-NH ₂	Amine	Methylamine
Halide	-X	Haloalkane	Chloroform (CHCl ₃)

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter F3b — IUPAC Nomenclature Basics

IUPAC (International Union of Pure and Applied Chemistry) naming lets you name any organic compound systematically. SSC CGL and RRB NTPC test the basic rules.

Suffix by hydrocarbon type

Series	Bond type	Suffix	Example (2 carbons)
Alkane	Single bonds only	-ane	Ethane (C_2H_6)

Series	Bond type	Suffix	Example (2 carbons)
Alkene	One double bond	-ene	Ethene (C ₂ H ₄)
Alkyne	One triple bond	-yne	Ethyne (C ₂ H ₂)

Prefix by carbon count

Carbons	Prefix	Example (alkane)
1	Meth-	Methane
2	Eth-	Ethane
3	Prop-	Propane
4	But-	Butane
5	Pent-	Pentane
6	Hex-	Hexane

Naming steps (in order)

1. **Find the longest continuous carbon chain** → gives the parent name prefix.
2. **Identify the principal functional group** → gives the suffix (-ane/-ene/-yne; -ol for alcohol; -al for aldehyde; -oic acid for carboxylic acid).
3. **Number the chain** from the end nearer to the principal functional group to get the **lowest locant**.
4. **Name substituents** (branches) alphabetically before the parent name, with their position numbers.

Example: CH₃-CH₂-OH → 2 carbons → "eth-" + alcohol suffix "-ol" + lowest locant "1" → **Ethan-1-ol** (ethanol).

KEY FACT

The three most important IUPAC rule facts for exams: (1) longest chain wins, (2) number to give functional group the lowest locant, (3) suffix determines compound class (-ane = alkane, -ene = alkene, -yne = alkyne, -ol = alcohol, -oic acid = carboxylic acid).

Chapter F4 — Fuels

EXAM ALERT

Fuel questions appear in almost every competitive exam. Know these three: **LPG = butane + propane; CNG = mainly methane; biogas = ~60% methane + CO₂**. The smell of LPG is due to ethyl mercaptan — LPG itself is odourless.

Composition of Common Fuels

Fuel	Main composition	Notes
LPG	Propane (C ₃ H ₈) + Butane (C ₄ H ₁₀)	Ethyl mercaptan added for smell detection
CNG	Methane (~85–95%) + small amounts of ethane, propane	Cleaner than petrol/diesel
Petrol (gasoline)	Hydrocarbons C ₅ –C ₁₂ (mainly octanes)	Octane number = % iso-octane
Diesel	Hydrocarbons C ₁₂ –C ₂₀	Cetane number = ignition quality
Kerosene	C ₁₂ –C ₁₅	Jet fuel, household
Biogas (gobar gas)	~60% CH ₄ + CO ₂ + traces H ₂ S	From animal waste fermentation

Fuel	Main composition	Notes
Coal gas	H ₂ (~50%) + CH ₄ (~30%) + CO	From destructive distillation of coal
Water gas	CO + H ₂ (1:1)	Coke + steam at high temp
Producer gas	CO + N ₂ (~70%)	Coke + air; cheaper but less energy
Natural gas	~85% methane + ethane + propane	Found with petroleum deposits

Calorific Values — Energy per kg

Fuel	kJ/kg (approximate)
Hydrogen	150,000
Methane (CNG)	55,000
LPG	50,000
Petrol / Diesel / Kerosene	45,000
Coal (anthracite)	33,000
Wood	17,000
Cow-dung cake	6,000

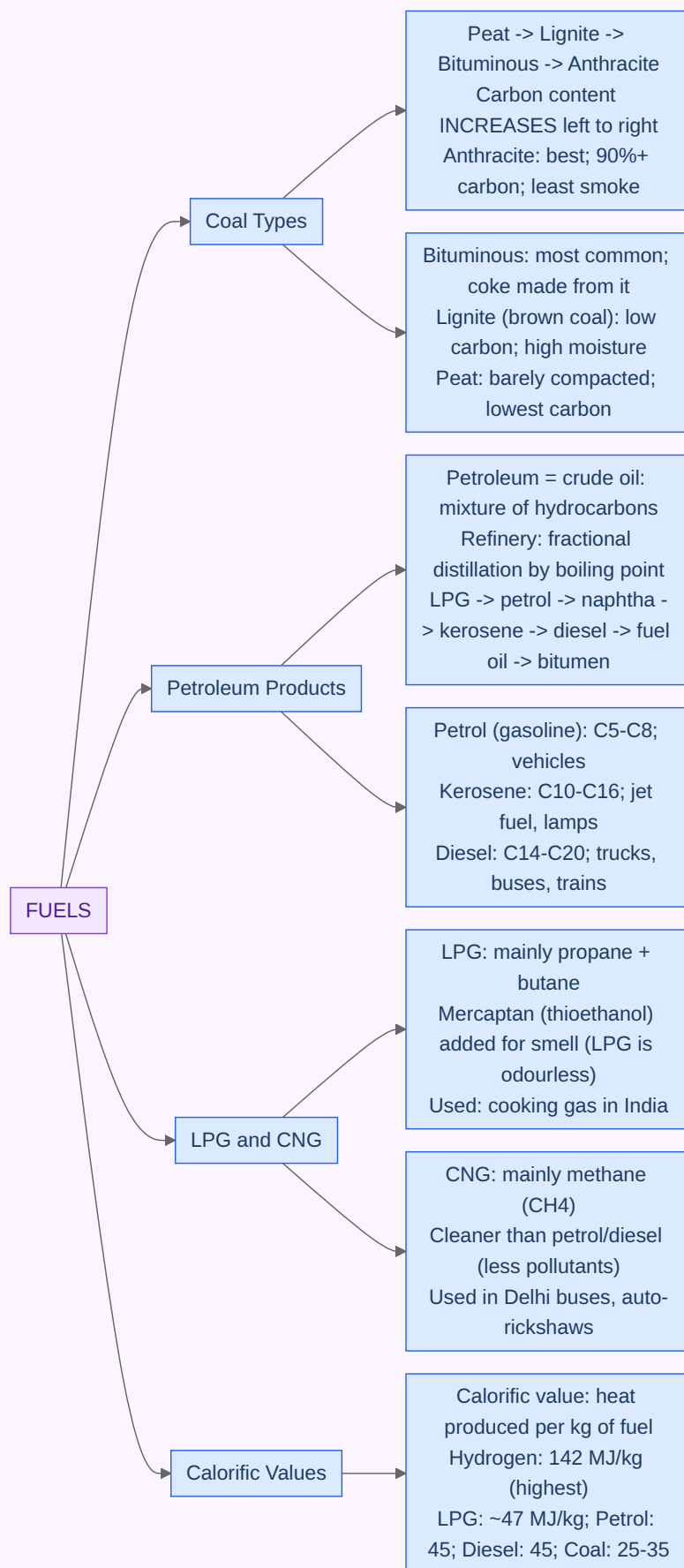
Hydrogen has by far the highest calorific value — nearly 3× petrol. This is why it is called the "fuel of the future."

Coal Types by Carbon Content

Type	% Carbon	Quality
Peat	50–60%	Lowest; youngest
Lignite (brown coal)	60–70%	Low
Bituminous	78–87%	Most widely used
Anthracite	87–95%	Highest CV; smokeless

PART G — POLYMERS AND FIBRES

CHAPTER SUMMARY — EVERYTHING IN ONE VIEW

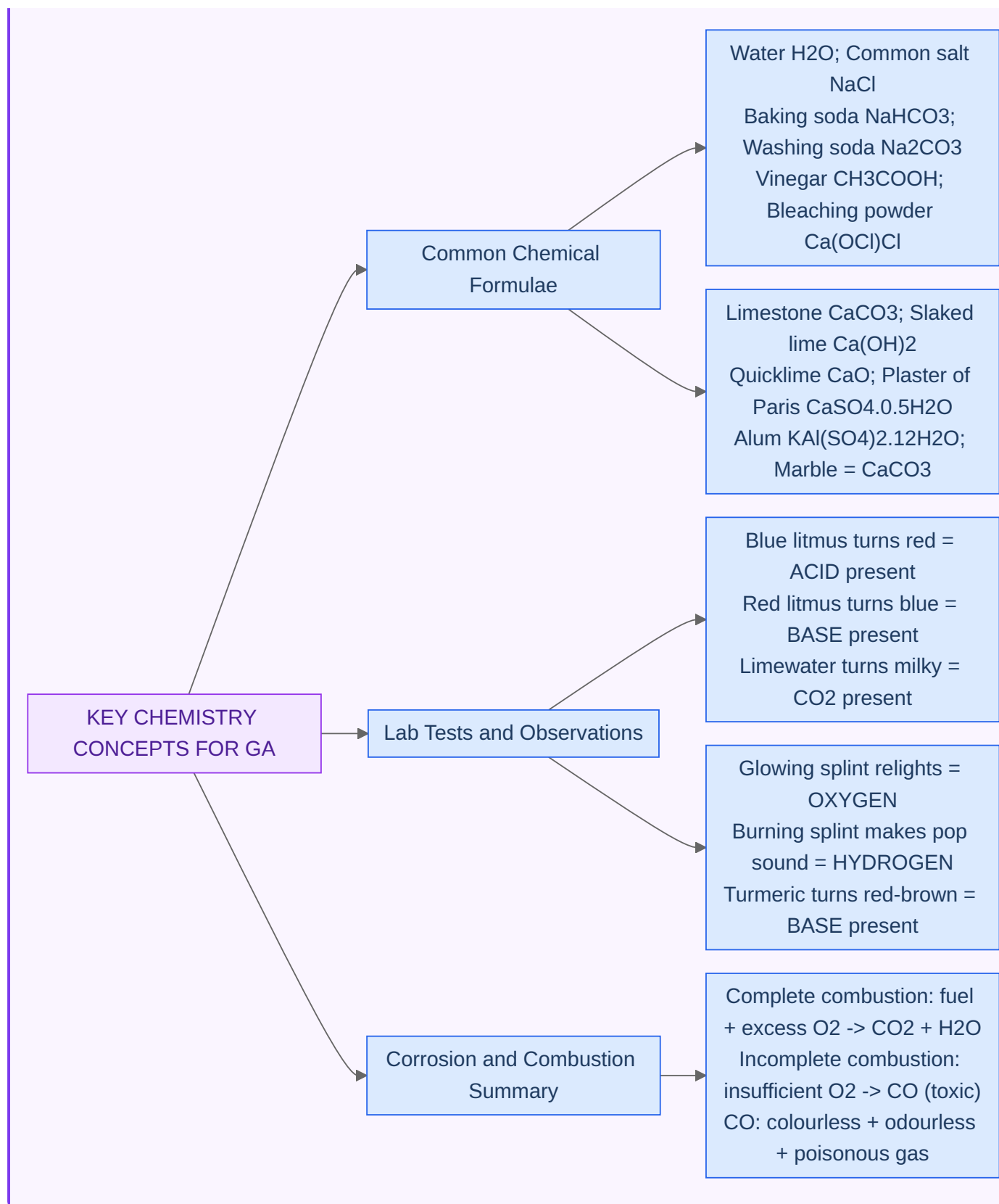


Chapter G1 — Key Concepts

- **Monomer** — small repeating unit.

- **Polymer** — long chain of monomers linked together.
- **Addition polymerisation** — monomers join without losing atoms (e.g., ethene → polythene).
- **Condensation polymerisation** — monomers join with release of a small molecule like water or HCl (e.g., nylon, polyester, bakelite).
- **Thermoplastic** — softens on heating, can be remoulded. (PVC, polythene, polypropylene)
- **Thermosetting** — once set, cannot be melted again. (Bakelite, melamine, epoxy resin)

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter G2 — The Important Polymers Table

Polymer	Monomer(s)	Type	Common Uses
Polythene (LDPE/HDPE)	Ethene	Addition, thermoplastic	Bags, bottles, pipes
Polypropylene (PP)	Propene	Addition, thermoplastic	Ropes, carpets, food containers
PVC	Vinyl chloride	Addition, thermoplastic	Pipes, raincoats, insulation

Polymer	Monomer(s)	Type	Common Uses
Polystyrene	Styrene	Addition, thermoplastic	Disposable cups, packaging foam
Teflon (PTFE)	Tetrafluoroethylene	Addition, thermoplastic	Non-stick cookware, lab seals
Plexiglas (PMMA)	Methyl methacrylate	Addition	Unbreakable glass, contact lenses
Nylon-6,6	Adipic acid + hexamethylenediamine	Condensation	Ropes, parachutes, stockings
Nylon-6	Caprolactam	Condensation	Carpets, fabric
Terylene / Dacron / PET	Ethylene glycol + terephthalic acid	Condensation	Polyester fabric, bottles, audio tape
Bakelite	Phenol + formaldehyde	Condensation, thermosetting	Electrical switches, handles
Melamine	Melamine + formaldehyde	Condensation, thermosetting	Crockery, fire-retardant fabrics
Buna-S (SBR)	Butadiene + styrene	Addition, synthetic rubber	Tyres, footwear
Buna-N (NBR)	Butadiene + acrylonitrile	Addition, synthetic rubber	Oil-resistant hoses, gaskets
Neoprene	Chloroprene	Addition, synthetic rubber	Wetsuits, weather-resistant seals
Natural rubber	Isoprene (polyisoprene)	Natural	Tyres (after vulcanisation)
Cellulose	β -glucose	Natural	Cotton, paper, wood
Starch	α -glucose	Natural	Food storage in plants
Rayon	Cellulose (treated)	Semi-synthetic	"Artificial silk"

EXAM ALERT

Three "first" questions that appear repeatedly: **First synthetic plastic = Bakelite (1907, Leo Baekeland)**. **First fully synthetic fibre = Nylon (1935, Wallace Carothers, DuPont)**. **Non-stick pan coating = Teflon (PTFE, Roy Plunkett, DuPont, 1938)**. Commit all three discoverers and years to memory.

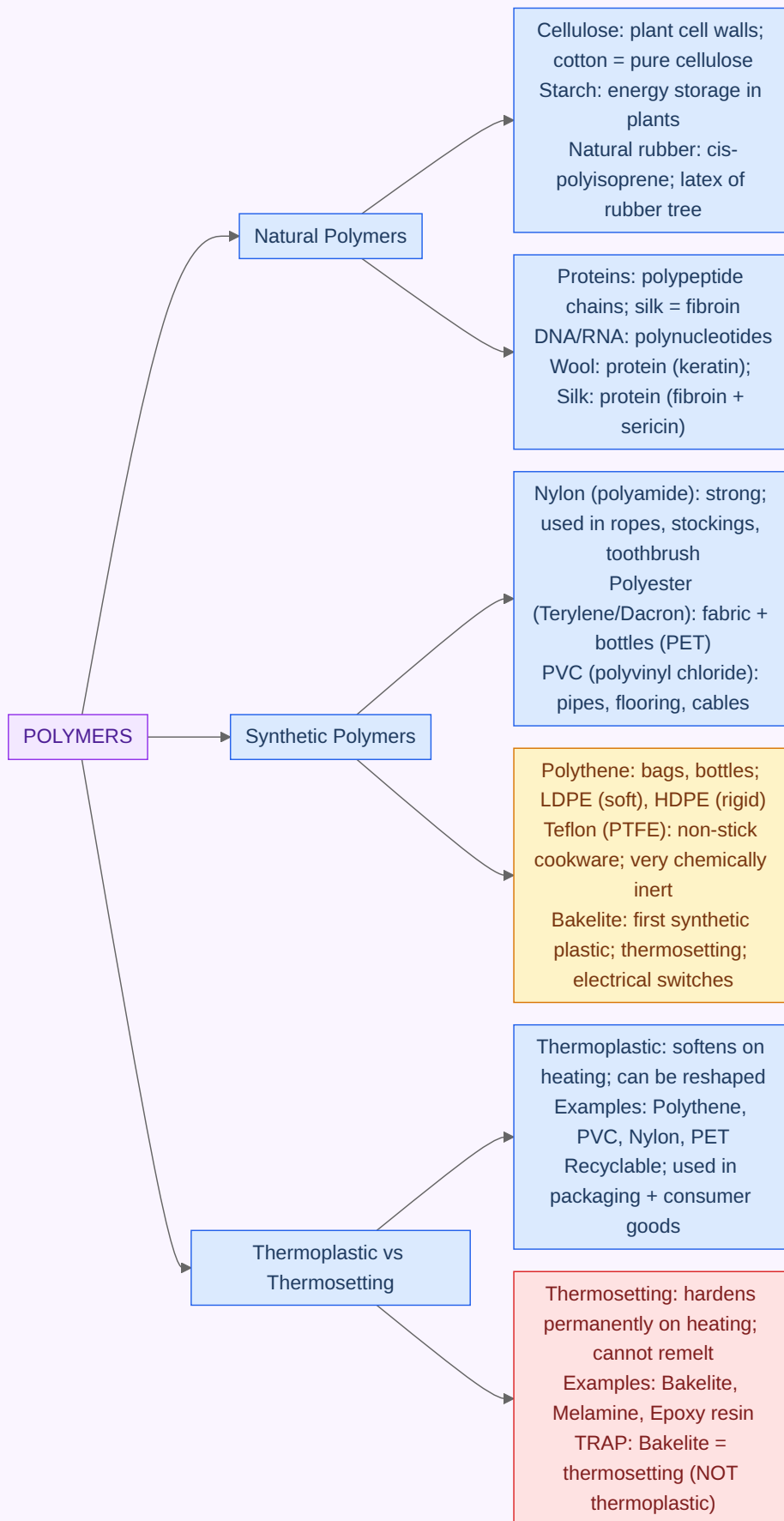
REMEMBER

Polymer quick links: - **Nylon** = Nylon ropes, stockings, parachutes (**Not** natural, **Nails** everything) - **Teflon** = Non-stick (Teflon = "Tef-**long** lasting non-stick") - **Bakelite** = Electrical switches — thermosetting (cannot melt again = "Bake once, set forever") - **PVC** = Pipes and wires (**P**ipes, **V**inyl, **C**ables — same 3 letters) - **PET/Terylene** = Bottles and polyester fabric (**P**lastic **E**very**T**ime = bottles and fabric)

EXAM ALERT

⚠ TRAP — Thermoplastic vs Thermosetting: "Bakelite can be remoulded by heating." — FALSE. Bakelite is **thermosetting** — once set, it cannot be melted again. Common thermoplastics (**CAN** remould): PVC, polythene, nylon, PET. Thermosetting (**CANNOT** remould): Bakelite, melamine, epoxy resin. Examiners ask this distinction specifically about Bakelite every few papers.

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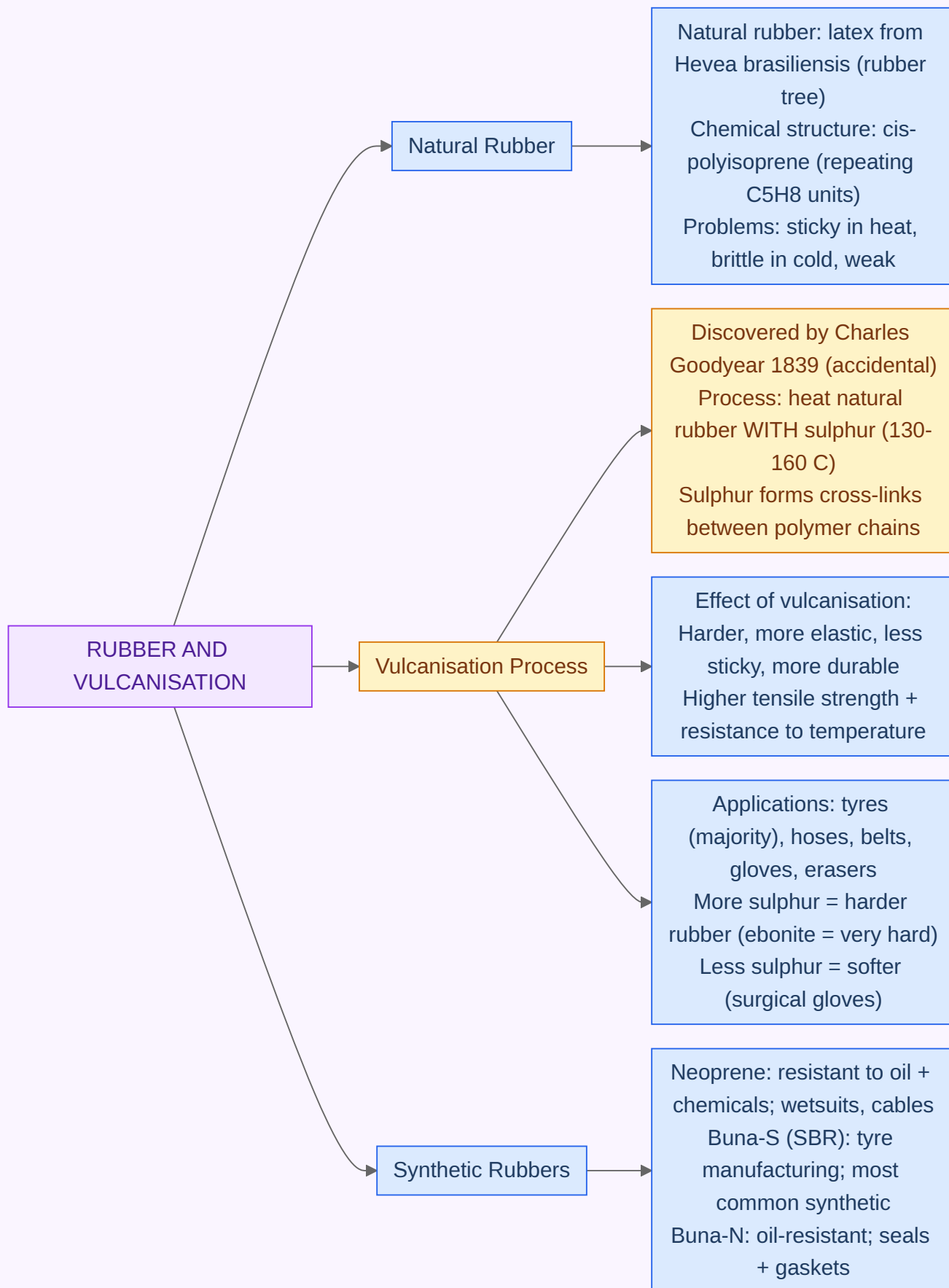


Chapter G3 — Vulcanisation of Rubber

Natural rubber (polyisoprene) is sticky, weak, and melts in summer heat. In 1839, Charles Goodyear discovered that heating raw rubber with sulphur creates cross-links between the polymer chains. This process — vulcanisation — makes rubber harder, stronger, elastic, and heat-stable. All tyre rubber is vulcanised.

PART H — EVERYDAY CHEMISTRY

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter H1 — Food Science

- **Fermentation** — yeast converts sugars into ethanol + CO₂ without oxygen (anaerobic). Basis of bread (CO₂ makes dough rise), beer, wine, idli.
- **Pasteurisation** — heating milk to 72 °C for 15 seconds, then rapid cooling. Kills pathogenic bacteria without changing flavour significantly. Developed by Louis Pasteur.
- **Hydrogenation** — liquid vegetable oils + H₂ gas (Ni catalyst) → solid fats (Vanaspati ghee). Alkenes become alkanes. Produces trans fats as a side product — these are associated with heart disease.
- **Iodised salt** — KIO₃ (potassium iodate) added to common salt to prevent iodine deficiency diseases, especially goitre.
- **Baking soda (NaHCO₃)** — on heating or in contact with acid, releases CO₂ which makes batter rise. Baking powder = NaHCO₃ + tartaric acid + starch.

Food Adulterants — Tested Questions

Food	Common adulterant	Detection test
Milk	Water, starch	Lactometer (for water); iodine turns blue (for starch)
Ghee	Vanaspati (vegetable fat)	Butyro - refractometer; Baudouin test
Honey	Sugar syrup	Fiehe's test (resorcinol + HCl)
Turmeric	Metanil yellow	Concentrated HCl → pink/red colour
Chilli powder	Brick powder	Water — brick powder settles
Mustard oil	Argemone oil	HNO ₃ turns yellow (sanguinarine test)
Pepper	Papaya seeds	Water — papaya seeds are lighter, float

Artificial Sweeteners — SSC Favourite

Artificial sweeteners are synthetic compounds that taste sweet but contribute zero or very few calories. SSC CGL tests the names, sweetness order, and who discovered them.

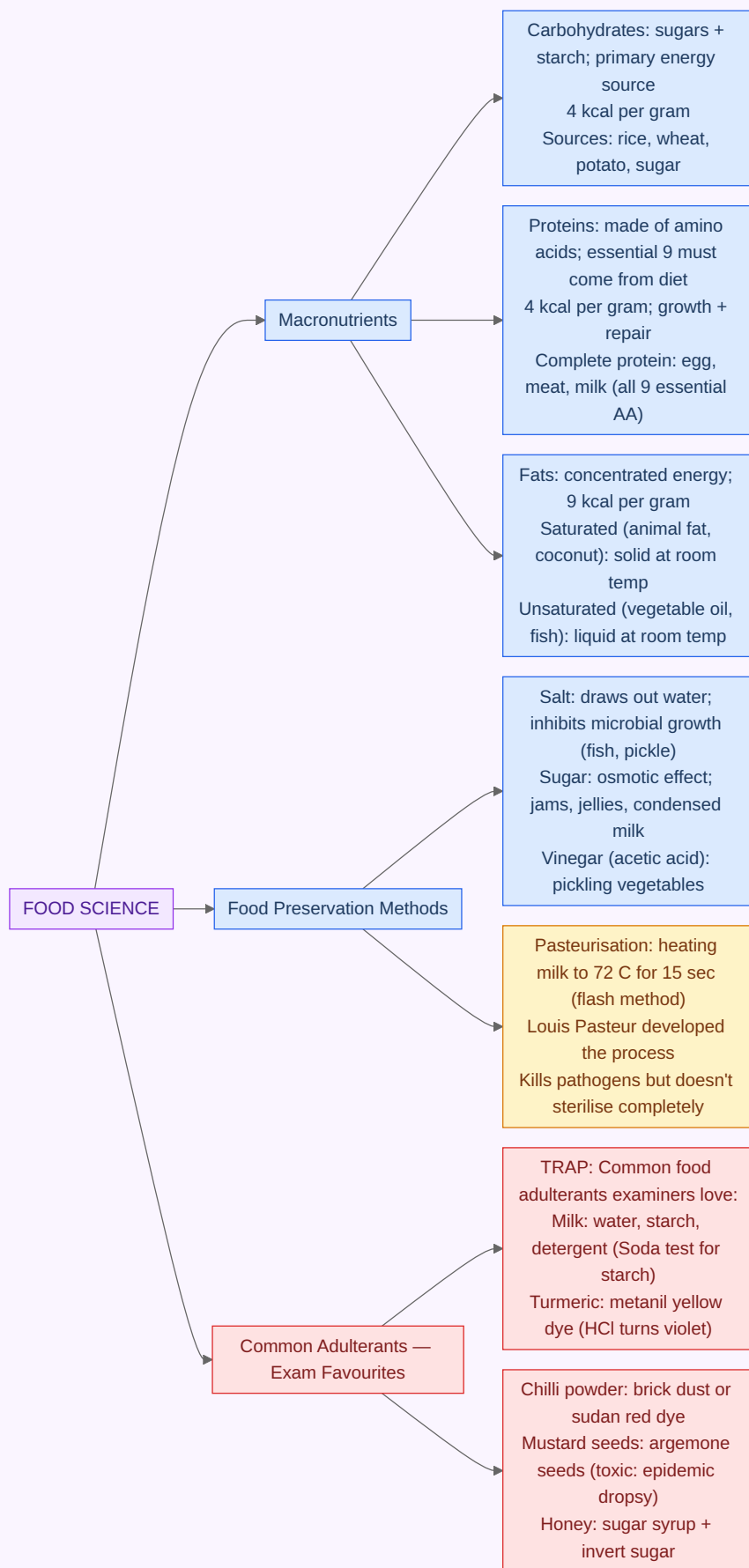
Sweetener	Sweetness vs sugar	Discovered	Key fact
Saccharin	~550 times	1879 (Fahlberg & Remsen)	First artificial sweetener; coal tar derivative; bitter aftertaste; banned in US 1977–2000 (ban later reversed)
Aspartame	~200 times	1965	Used in diet colas (Pepsi Max, Diet Coke); breaks down at high temperature; NOT used in baking
Sucralose	~600 times	1976	Made from sugar (chlorinated sucrose); heat-stable; used in baking products
Stevia	~200–300 times	—	Natural sweetener from Stevia plant leaves; widely used in India

KEY FACT

Sweetness order (high-yield): Sucralose (600×) > Saccharin (550×) > Aspartame (200×) > Sugar (1×).
Saccharin = **FIRST** artificial sweetener (1879). Sucralose = **SWEETEST** of the three most common ones.

COMMON TRAP

Exam trap: "Saccharin is derived from sugar" — **FALSE**. Saccharin is derived from **toluene/coal tar** (it is synthetic). Sucralose **IS** derived from sugar (chlorinated sucrose). Examiners swap these descriptions.



Chapter H2 — Soaps, Detergents, Water

Soap

Soap = sodium salt (or potassium salt) of a long-chain fatty acid. Made by saponification — heating fat/oil with NaOH. The soap molecule has a long hydrophobic tail (dissolves in oil) and a hydrophilic ionic head (dissolves in water). This is how it removes grease: the tails embed in the oil droplet, the heads stick out into water.

Soap does NOT work well in hard water. Hard water contains dissolved Ca^{2+} and Mg^{2+} ions. These react with soap to form insoluble scum (calcium stearate), which wastes soap and leaves a film.

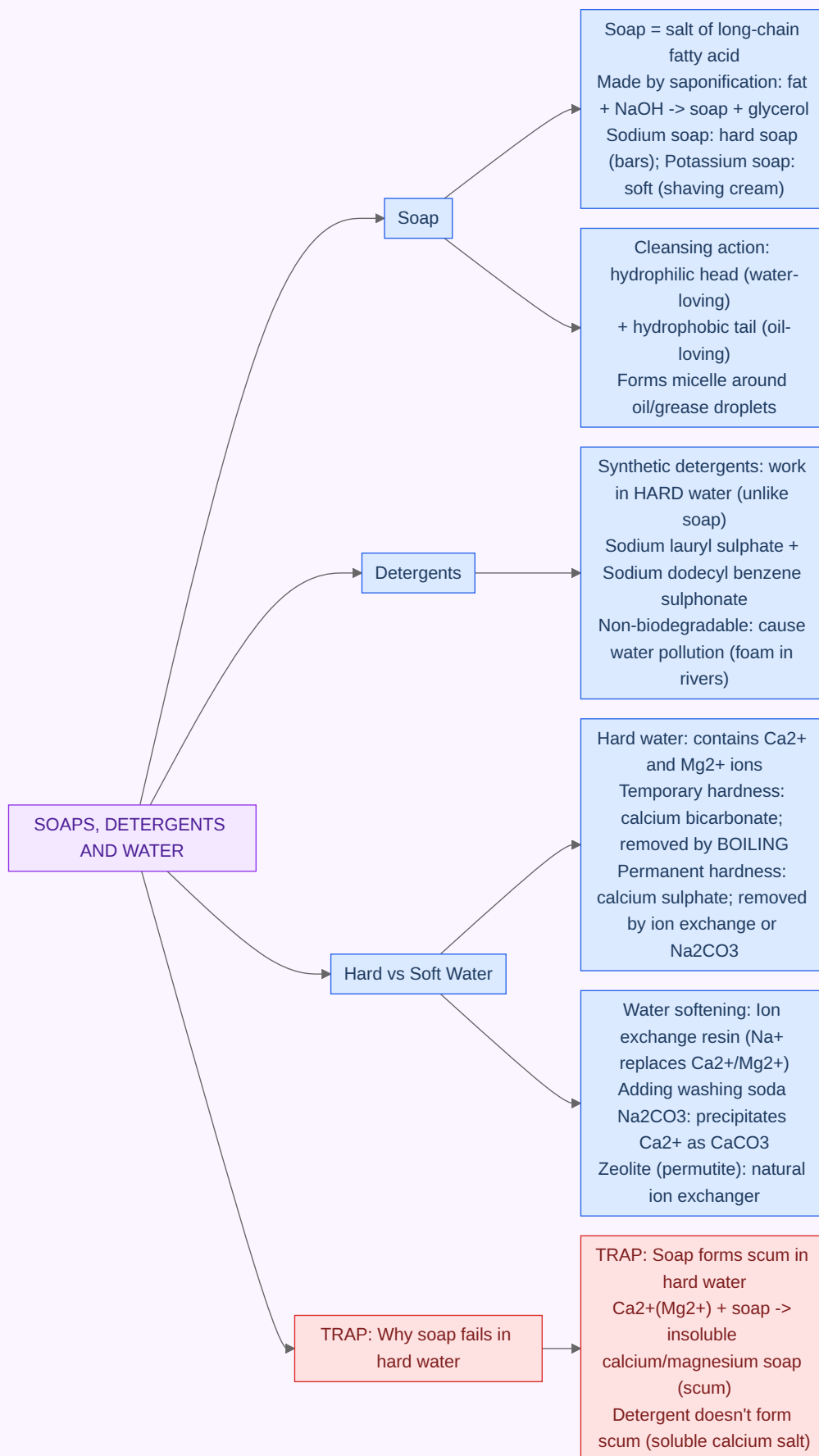
Detergent

Synthetic detergents (alkyl benzene sulphonates) work in hard water because their calcium/magnesium salts are soluble. Newer detergents are biodegradable; older ones (branched-chain alkyl sulphonates) were not.

Hard Water

- **Temporary hardness** — caused by dissolved $\text{Ca}(\text{HCO}_3)_2$ and $\text{Mg}(\text{HCO}_3)_2$. Removed by boiling (bicarbonates decompose to insoluble carbonates that precipitate).
- **Permanent hardness** — caused by CaSO_4 , MgSO_4 , CaCl_2 . Not removed by boiling. Removed by ion-exchange resins (zeolite) or adding Na_2CO_3 (washing soda).

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Chapter H3 — Medicines

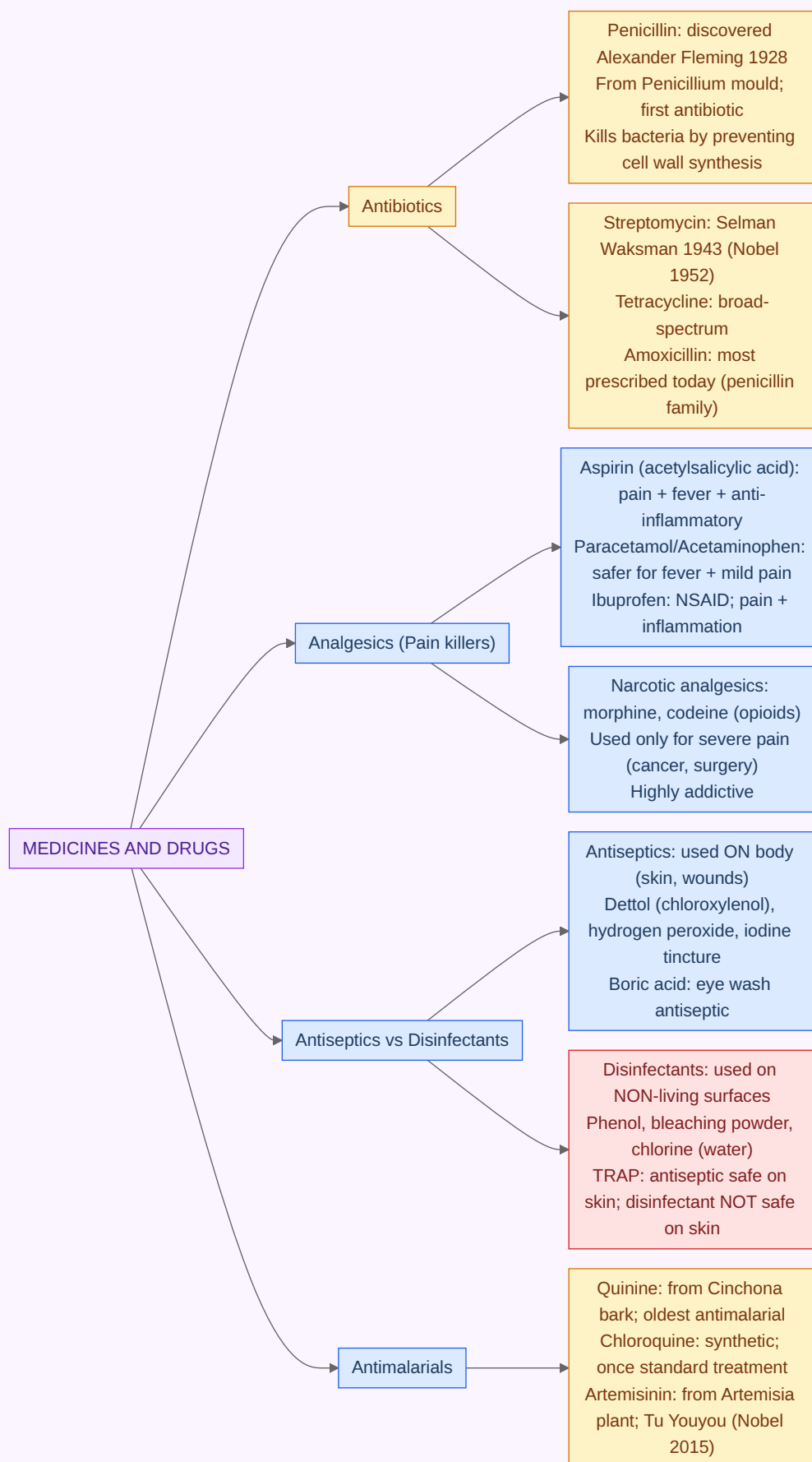
Drug class	What it does	Examples
Analgesic (painkiller)	Reduces pain	Aspirin, paracetamol, ibuprofen
Antipyretic	Reduces fever	Paracetamol (also analgesic)
Antibiotic	Kills bacteria	Penicillin, amoxicillin, streptomycin, tetracycline
Antiseptic	Kills microbes on living tissue	Dettol, iodine, hydrogen peroxide, savlon
Disinfectant	Kills microbes on non-living surfaces	Phenol, formalin, bleaching powder
Antacid	Neutralises stomach acid	Sodium bicarbonate, milk of magnesia, gelusil
Antihistamine	Treats allergic reactions	Cetirizine, chlorpheniramine
Antimalarial	Treats malaria	Chloroquine, quinine, artemisinin
Antitubercular	Treats TB	Rifampicin, isoniazid, pyrazinamide, ethambutol
Tranquilliser	Reduces anxiety	Diazepam (Valium), equanil

KEY FACT

Penicillin — discovered by Alexander Fleming in **1928** when he noticed the mould *Penicillium notatum* killing bacteria in a petri dish. This remains one of the most frequently tested discovery facts in chemistry/general science.

PART I — ENVIRONMENTAL CHEMISTRY

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Chapter I1 — Air Pollution

Pollutant	Source	Effect
CO (carbon monoxide)	Incomplete combustion, vehicle exhaust	Binds haemoglobin 240× stronger than O ₂ → suffocation
SO ₂ (sulphur dioxide)	Coal combustion → combines with rain → H ₂ SO ₄	Acid rain; respiratory irritant
NO _x (nitrogen oxides)	Vehicle exhaust, thunderstorms	Photochemical smog; acid rain
PM _{2.5} / PM ₁₀	Dust, smoke, vehicle exhaust	Deep lung penetration; respiratory disease
Ground-level O ₃	Photochemical reaction of NO _x + VOCs in sunlight	Lung irritant; reduces crop yields
Greenhouse gases	Various sources	Global warming

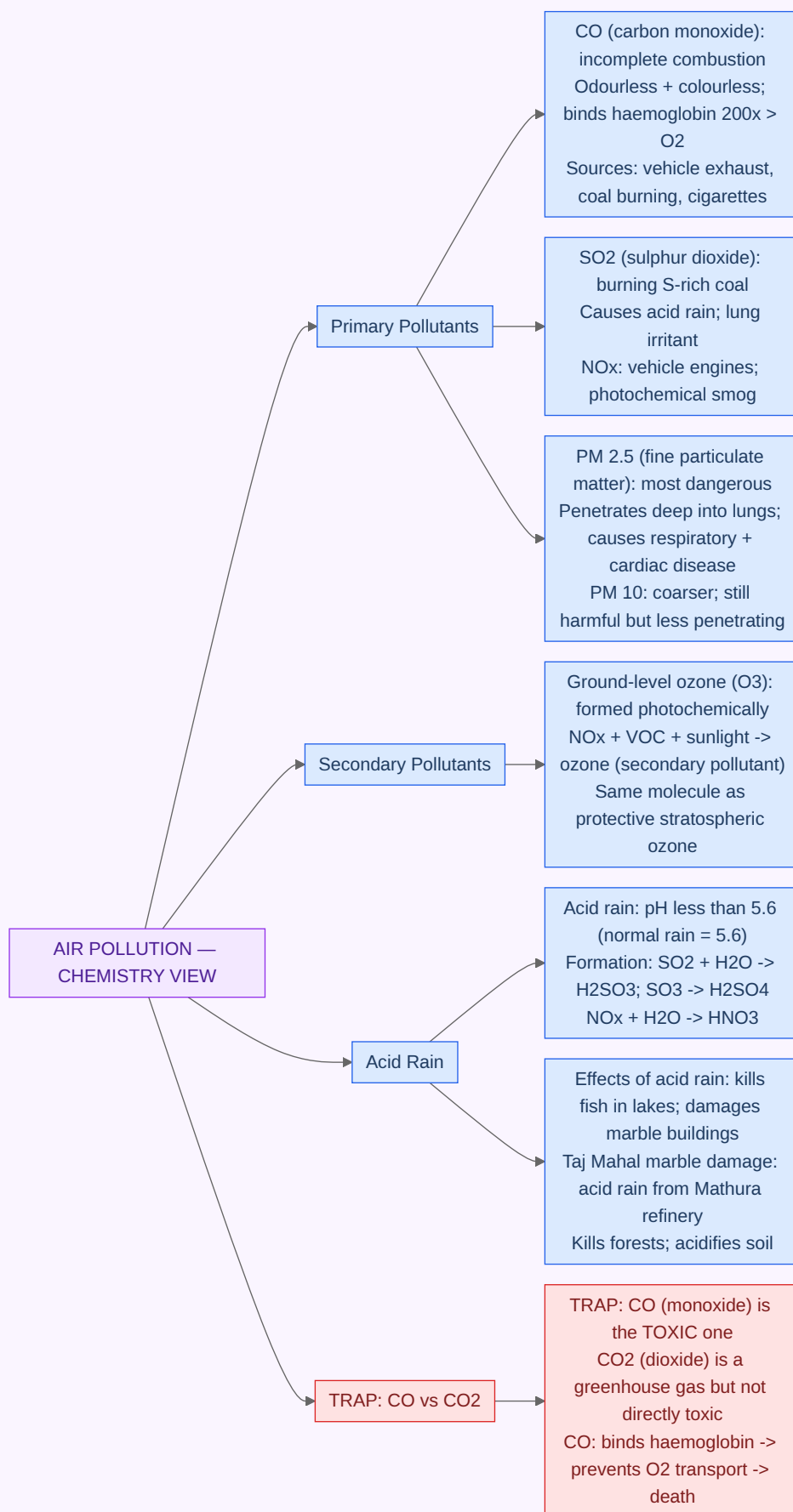
Greenhouse Gases — Global Warming Potential

Gas	GWP (relative to CO ₂)
CO ₂	1 (reference)
CH ₄ (methane)	28–36
N ₂ O	298
HFCs	Hundreds to thousands
SF ₆	23,500

EXAM ALERT

The main greenhouse gas by volume is CO₂. But the strongest greenhouse gas by GWP is SF₆ (sulphur hexafluoride). Examiners ask both. "Most abundant greenhouse gas in the atmosphere" = **water vapour (H₂O)** — but examiners almost always mean CO₂ when they ask this, as CO₂ is the main anthropogenic one.

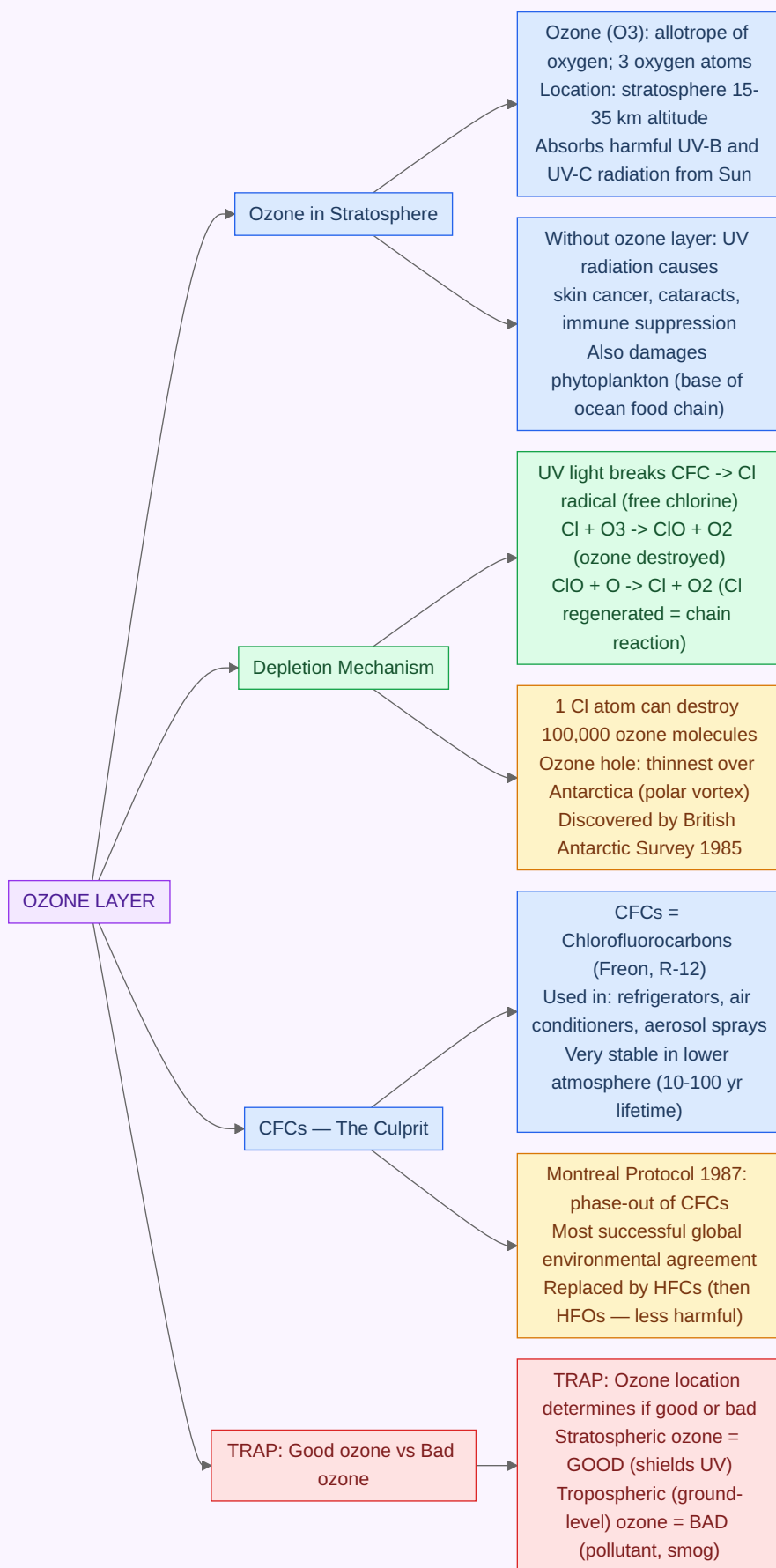
⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter 12 — Ozone Layer

- Ozone (O_3) in the stratosphere (20–30 km altitude) absorbs harmful UV-B and UV-C radiation from the Sun.
- CFCs (chlorofluorocarbons) — used in refrigerants and aerosols — decompose in the stratosphere and release chlorine atoms that catalytically destroy ozone.
- The ozone hole over Antarctica was discovered in 1985.
- **Montreal Protocol (1987)** — international treaty phasing out CFCs. Most successful environmental treaty.
- **Kigali Amendment (2016)** — extends Montreal Protocol to phase out HFCs (hydrofluorocarbons), which replaced CFCs but are potent greenhouse gases.

CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter 13 — Water Pollution

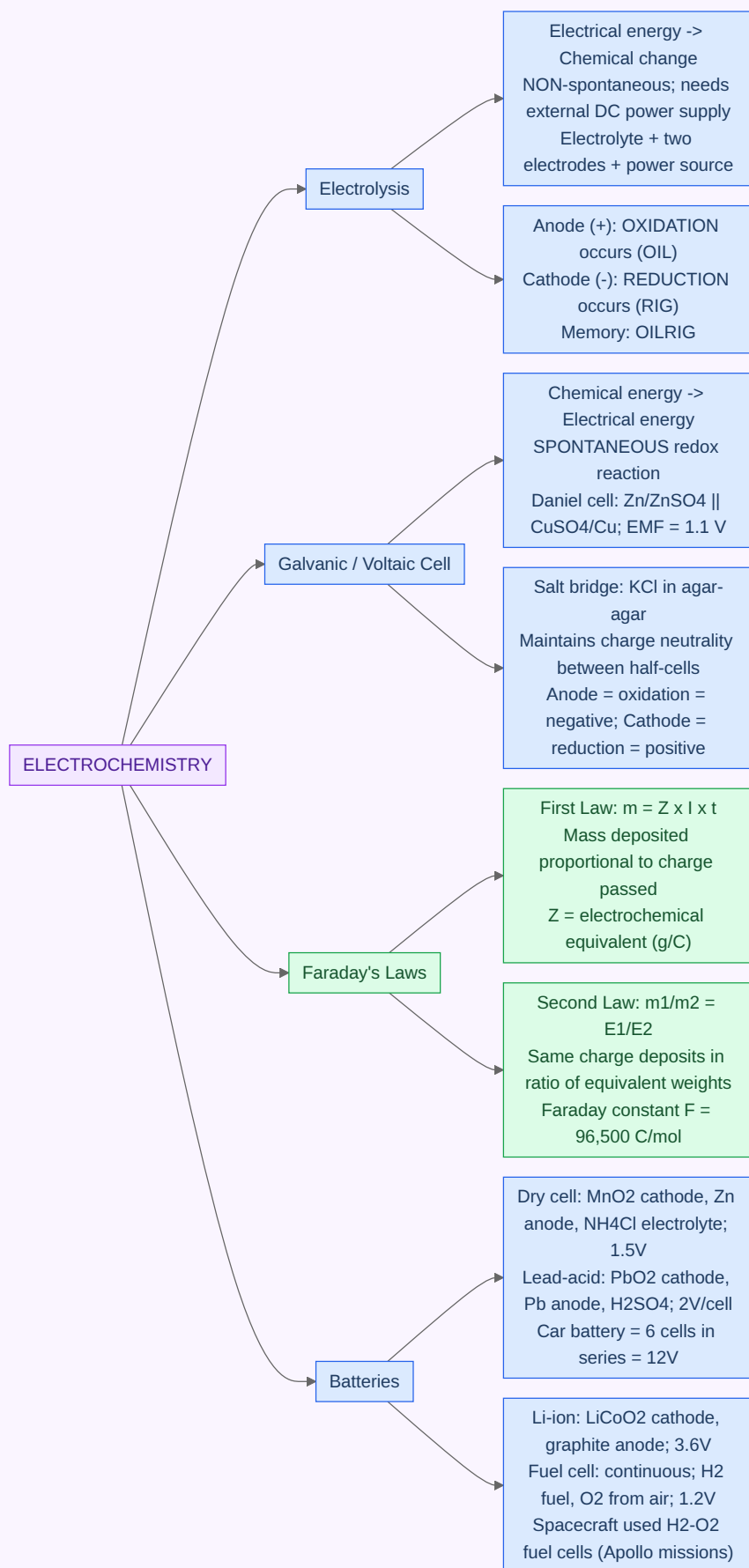
- **BOD (Biological Oxygen Demand)** — amount of oxygen needed by bacteria to decompose organic matter in water. Higher BOD = more pollution.
- **DO (Dissolved Oxygen)** — fish need at least 4 mg/L. Heavily polluted water can have DO near zero.
- **Eutrophication** — excess nutrients (nitrates, phosphates from fertiliser runoff) → algal bloom → algae die → bacteria decompose them using up all oxygen → fish die.
- **Fluorosis** — fluoride >1.5 ppm in drinking water → mottled teeth, skeletal fluorosis.
- **Blue baby syndrome (Methaemoglobinaemia)** — high nitrates in well water → converted to nitrites by gut bacteria → blocks oxygen in infant blood.

Famous Pollution Disasters

Disaster	Cause	Location
Minamata disease	Mercury (Hg) poisoning from industrial discharge	Minamata Bay, Japan (1950s)
Itai-Itai disease	Cadmium (Cd) poisoning from mining discharge	Jinzu River, Japan (1950s)
Bhopal gas tragedy	MIC (methyl isocyanate) gas leak	Bhopal, India, 1984

PART J — ELECTROCHEMISTRY

⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter J1 — What is Electrochemistry?

Electrochemistry is the branch of chemistry dealing with the interconversion of **electrical energy** and **chemical energy**. It covers two fundamental processes:

- **Electrolysis** — electrical energy drives a non-spontaneous chemical change (external power supply forces a reaction that wouldn't occur on its own).
- **Galvanic/Voltaic cell** — a spontaneous redox reaction produces electrical energy (chemical reaction generates a current).

KEY FACT

The key distinction: **Electrolysis = electricity IN → chemistry OUT** (non-spontaneous). **Galvanic cell = chemistry IN → electricity OUT** (spontaneous). Examiners regularly swap these — know which direction the energy flows.

Chapter J2 — Electrolytic Cell

An electrolytic cell consists of: 1. An **electrolyte** — an ionic compound in solution or molten form (conducts electricity via ion movement). 2. Two **electrodes** — conductors dipped into the electrolyte. 3. An **external DC power supply** — forces the non-spontaneous reaction.

Electrode Rules (very frequently tested)

Electrode	Terminal	Charge	Reaction	Memory
Anode	+ve terminal	Positive	Oxidation (loses electrons)	OIL
Cathode	-ve terminal	Negative	Reduction (gains electrons)	RIG

KEY FACT

OILRIG — Oxidation Is Loss, Reduction Is Gain. This applies in BOTH electrolytic cells AND galvanic cells. The anode is always where oxidation occurs; the cathode is always where reduction occurs.

Key Electrolysis Examples (exam-critical table)

Electrolyte	Cathode product	Anode product	Industrial use
Dilute H_2SO_4 (dil. sulphuric acid)	H_2 gas	O_2 gas	Production of hydrogen
CuSO_4 solution (copper electrodes)	Cu deposited on cathode	Cu dissolved from anode	Electroplating / copper refining
Molten NaCl — Down's process	Na metal	Cl_2 gas	Industrial sodium metal production
Molten Al_2O_3 in cryolite — Hall-Hérout process	Al metal	O_2 (burns carbon anode)	Industrial aluminium extraction
Brine (NaCl solution) — Chlor-alkali process	H_2 gas	Cl_2 gas; NaOH formed in solution	Bleaching powder, PVC, caustic soda

COMMON TRAP

Exam trap — electrolysis of dilute H_2SO_4 : H_2 forms at the **CATHODE** (negative electrode) and O_2 forms at the **ANODE** (positive electrode). Students frequently reverse this. Remember: H^+ ions move to the negative cathode → H_2 forms there. $\text{O}^{2-}/\text{OH}^-$ ions move to positive anode → O_2 forms there.

EXAM ALERT

Electroplating uses CuSO_4 solution: the object to be plated is the **cathode** (Cu deposits on it); the copper bar is the **anode** (Cu dissolves to replenish Cu^{2+} ions). The electrolyte concentration stays constant throughout the process.

Chapter J3 — Faraday's Laws of Electrolysis (VERY high yield)

Michael Faraday (1834) gave two quantitative laws governing electrolysis.

First Law

The mass of substance deposited at an electrode is directly proportional to the quantity of electricity (charge) passed.

$$m = Z \times Q = Z \times I \times t$$

Where: - m = mass deposited (grams) - Z = **electrochemical equivalent** (g per coulomb) - Q = charge passed (coulombs) = $I \times t$ - I = current (amperes) - t = time (seconds)

Second Law

When the same quantity of electricity is passed through different electrolytes, the masses of substances deposited are proportional to their equivalent weights.

$$\frac{m_1}{m_2} = \frac{E_1}{E_2}$$

Where E = equivalent weight = Atomic mass $\div$ Valency.

Example: Same charge deposits 1.08 g of Ag ($E = 108$) and 0.32 g of Cu ($E = 32$). Ratio = $108 : 32 = 27 : 8$. ✓

The Faraday Constant

KEY FACT

1 Faraday (F) = 96,500 coulombs per mole = charge carried by one mole of electrons (6.022×10^{23} electrons). This is the single most important number in electrochemistry for competitive exams. $F = 96,485 \text{ C/mol}$ exactly, but use 96,500 in numerical problems.

Electrochemical equivalent: $Z = \text{Equivalent weight} / 96,500 \text{ (g/C)}$

Quick formula to remember:

$$m = \frac{E \times I \times t}{96500} = \frac{E \times Q}{96500}$$

Chapter J4 — Galvanic / Voltaic Cell

A galvanic cell converts a **spontaneous** chemical (redox) reaction into electrical energy. It consists of: - Two **half-cells** (each contains a metal electrode in its salt solution) - A **salt bridge** — a U-tube containing KCl in

agar-agar gel; allows ion flow to maintain electrical neutrality without mixing the two solutions - An **external wire** — carries electron flow (the actual current)

The Daniel Cell (most exam-tested example)

Zn | ZnSO₄ (aq) || CuSO₄ (aq) | Cu

Half-cell	Electrode	Process	Sign
Zinc in ZnSO ₄	$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$	Oxidation	Anode (-ve terminal)
Copper in CuSO ₄	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	Reduction	Cathode (+ve terminal)

- **Zinc plate dissolves** over time (Zn is oxidised, ions go into solution).
- **Copper deposits** on the copper plate (Cu²⁺ ions from solution are reduced).
- **EMF ≈ 1.1 V**
- Electrons flow externally from Zn (anode) → Cu (cathode).
- In the salt bridge, K⁺ ions move toward the CuSO₄ half-cell, Cl⁻ ions move toward the ZnSO₄ half-cell — maintaining neutrality.

Cell notation rule: Anode written LEFT | anode solution || cathode solution | Cathode RIGHT (double bar = salt bridge).

EMF of a cell

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

For the Daniel cell: $E^\circ = +0.34 \text{ V} - (-0.76 \text{ V}) = \mathbf{+1.10 \text{ V}}$

A positive E°_{cell} means the reaction is spontaneous.

Chapter J5 — Standard Electrode Potential and Electrochemical Series

Every electrode has a standard electrode potential (E°) measured against the **Standard Hydrogen Electrode (SHE)**, which is set at **0.00 V**.

Reading the series

Half-reaction (reduction)	E° (V)	Notes
$\text{F}_2 + 2\text{e}^- \rightarrow 2\text{F}^-$	+2.87	Strongest oxidising agent
$\text{Au}^{3+} + 3\text{e}^- \rightarrow \text{Au}$	+1.50	Noble metal
$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$	+1.36	
$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	+0.34	
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	0.00	Reference (SHE)
$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$	-0.44	
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$	-0.76	
$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$	-2.71	
$\text{Li}^+ + \text{e}^- \rightarrow \text{Li}$	-3.04	Strongest reducing agent

KEY FACT

Higher E° → stronger oxidising agent → tends to be reduced (cathode side). Lower E° (more negative) → stronger reducing agent → tends to be oxidised (anode side). The electrochemical series also predicts which metal displaces another: a metal lower in the series (more negative E°) can displace a metal higher

in the series (less negative / positive E°) from its salt solution. This links directly to the activity series (Part E).

Chapter J6 — Batteries (Common Exam Questions)

A **battery** is one or more galvanic cells connected in series. - **Primary (non-rechargeable)** — reaction is irreversible; cannot be recharged. - **Secondary (rechargeable)** — electrode reactions are reversible; can be charged and reused. - **Fuel cell** — reactants (fuel + oxidant) supplied continuously; produces electricity as long as fuel flows.

Battery Comparison Table

Battery	Primary / Secondary	Cathode	Anode	Electrolyte	EMF	Common use
Dry cell (Leclanché)	Primary	MnO ₂	Zn	NH ₄ Cl paste	1.5 V	Torches, remotes, clocks
Lead-acid	Secondary	PbO ₂	Pb	Dilute H ₂ SO ₄	2 V/cell	Car batteries (12 V = 6 cells)
Nickel-Cadmium (NiCd)	Secondary	NiO(OH)	Cd	KOH	1.2 V	Cordless power tools
Lithium-ion (Li-ion)	Secondary	LiCoO ₂	Graphite	Li salt in organic solvent	3.6 V	Smartphones, EVs, laptops
H ₂ -O ₂ Fuel cell	Fuel cell (continuous)	O ₂ (air)	H ₂	KOH / PEM membrane	~1.2 V	Spacecraft, EVs, buses

COMMON TRAP

Trap cards — batteries:

- "Car battery uses which acid?" → **Dilute H₂SO₄** (sulphuric acid) — NOT HCl, NOT HNO₃.
- "Which battery is used in spacecraft / Apollo missions?" → **H₂-O₂ Fuel cell** (also produces drinking water as a by-product).
- "Dry cell electrolyte?" → **NH₄Cl** (ammonium chloride paste) — NOT sulphuric acid.
- "What is the EMF of a single lead-acid cell?" → **2 V**. A 12 V car battery = 6 cells in series.
- "Li-ion battery cathode?" → **LiCoO₂** (lithium cobalt oxide). Anode = graphite (intercalation electrode).
- "Faraday constant value?" → **96,500 C/mol** (or 96,485 C/mol exactly).
- "In a galvanic cell, which electrode is the anode?" → The one where **oxidation** occurs — the **negative** terminal (zinc in the Daniel cell). In electrolytic cells the anode is +ve (connected to the +ve terminal of the external supply).

Chapter J — Self-test: 5 PYQ-style questions

- Which of the following laws states that mass deposited during electrolysis is proportional to the quantity of electricity passed? (*Faraday's First Law*)
- In the electrolysis of dilute H₂SO₄, which gas is produced at the cathode? (*H₂*)
- The EMF of a Daniel cell is approximately:
(a) 0.34 V (b) 0.76 V **(c) 1.10 V** (d) 2.00 V
- Which battery is used in Apollo spacecraft? (*H₂-O₂ Fuel cell*)
- 1 Faraday of charge = how many coulombs? (*96,500 C*)

PART T — DISCOVERIES AND INVENTORS IN CHEMISTRY

Discovery + year questions appear in every SSC and banking paper — learn each entry here.

Discovery / Invention	Person	Year
Atomic theory	John Dalton	1808
Electron	J.J. Thomson	1897
Radioactivity	Henri Becquerel	1896
Polonium + Radium	Marie & Pierre Curie	1898
Alpha, beta particles	Ernest Rutherford	1899
Nuclear model of atom	Ernest Rutherford	1911
Proton	Ernest Rutherford	1919
Neutron	James Chadwick	1932
Modern periodic table (by atomic number)	Henry Moseley	1913
Periodic law (by atomic weight)	Dmitri Mendeleev	1869
Synthesis of urea from inorganic compounds	Friedrich Wöhler	1828
Hydrogen	Henry Cavendish	1766
Oxygen	Joseph Priestley + Carl Scheele	1774
Nitrogen	Daniel Rutherford	1772
Vulcanisation of rubber	Charles Goodyear	1839
First synthetic dye (mauveine)	William Perkin	1856
Dynamite	Alfred Nobel	1867
Rayon (artificial silk)	Hilaire de Chardonnet	1884
Penicillin	Alexander Fleming	1928
Streptomycin (first anti - TB antibiotic)	Selman Waksman	1943
Bakelite (first true synthetic plastic)	Leo Baekeland	1907
Polythene	Eric Fawcett + Reginald Gibson (ICI)	1933
Nylon (first fully synthetic fibre)	Wallace Carothers, DuPont	1935
Teflon (PTFE)	Roy Plunkett, DuPont	1938
Haber process (ammonia synthesis)	Fritz Haber + Carl Bosch	1909–1913
Contact process (sulphuric acid)	Peregrine Phillips	1831
Ostwald process (nitric acid)	Wilhelm Ostwald	1902
Hall - Héroult process (aluminium)	Charles Hall + Paul Héroult	1886
DDT (use as insecticide)	Paul Müller	1939
CFC refrigerants	Thomas Midgley Jr	1928
TNT	Julius Wilbrand	1863

PART U — THE ESSENTIAL CHEMICAL NAMES TABLE

This is the table you must memorise completely. Chemical name ↔ everyday name questions appear in every competitive exam paper.

Everyday Name	Chemical Name	Formula
Common salt	Sodium chloride	NaCl
Baking soda	Sodium bicarbonate	NaHCO ₃
Washing soda	Sodium carbonate decahydrate	Na ₂ CO ₃ ·10H ₂ O
Caustic soda	Sodium hydroxide	NaOH
Caustic potash	Potassium hydroxide	KOH
Quicklime	Calcium oxide	CaO
Slaked lime	Calcium hydroxide	Ca(OH) ₂
Limestone / Marble / Chalk	Calcium carbonate	CaCO ₃
Plaster of Paris (POP)	Calcium sulphate hemihydrate	CaSO ₄ ·½H ₂ O
Gypsum	Calcium sulphate dihydrate	CaSO ₄ ·2H ₂ O
Bleaching powder	Calcium hypochlorite (oxychloride)	CaOCl ₂
Hypo (photography fixer)	Sodium thiosulphate	Na ₂ S ₂ O ₃ ·5H ₂ O
Milk of magnesia	Magnesium hydroxide	Mg(OH) ₂
Epsom salt	Magnesium sulphate heptahydrate	MgSO ₄ ·7H ₂ O
Glauber's salt	Sodium sulphate decahydrate	Na ₂ SO ₄ ·10H ₂ O
Borax	Sodium tetraborate decahydrate	Na ₂ B ₄ O ₇ ·10H ₂ O
Alum	Potassium aluminium sulphate	KAl(SO ₄) ₂ ·12H ₂ O
Mohr's salt	Ferrous ammonium sulphate	(NH ₄) ₂ Fe(SO ₄) ₂ ·6H ₂ O
Blue vitriol	Copper sulphate pentahydrate	CuSO ₄ ·5H ₂ O
Green vitriol	Ferrous sulphate heptahydrate	FeSO ₄ ·7H ₂ O
White vitriol	Zinc sulphate heptahydrate	ZnSO ₄ ·7H ₂ O
Red lead	Lead tetroxide	Pb ₃ O ₄
Litharge	Lead monoxide	PbO
Dry ice	Solid carbon dioxide	CO ₂ (s)
Heavy water	Deuterium oxide	D ₂ O
Vinegar	Dilute acetic acid	CH ₃ COOH (5–8%)
Spirit / Rectified spirit	Ethanol	C ₂ H ₅ OH
Wood spirit (poison)	Methanol	CH ₃ OH
Marsh gas	Methane	CH ₄
Laughing gas	Nitrous oxide	N ₂ O
Tear gas	Chloroacetophenone / CS gas	Various
Mustard gas	Dichlorodiethyl sulphide	(ClCH ₂ CH ₂) ₂ S
Aqua regia	3 HCl + 1 HNO ₃	— (dissolves Au, Pt)
Aqua fortis	Concentrated nitric acid	HNO ₃
Oil of vitriol	Concentrated sulphuric acid	H ₂ SO ₄
Spirit of salt	Hydrochloric acid	HCl
Sugar of lead	Lead acetate	Pb(CH ₃ COO) ₂

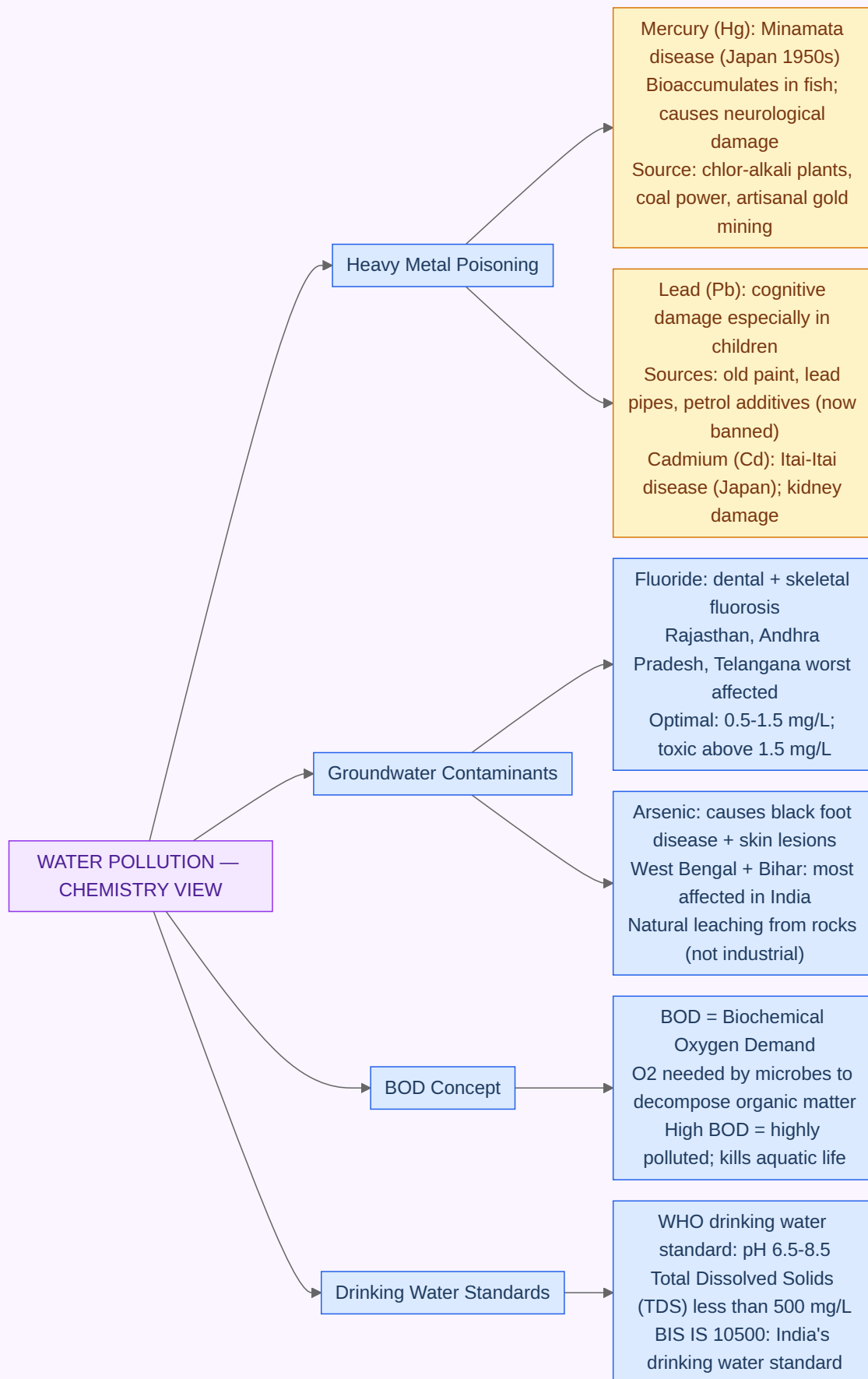
Everyday Name	Chemical Name	Formula
Vermillion / Sindoor	Mercuric sulphide	HgS
Galena	Lead sulphide	PbS
Cinnabar	Mercuric sulphide	HgS
Copperas	Ferrous sulphate	FeSO ₄ ·7H ₂ O
Producer gas	Carbon monoxide + Nitrogen	CO + N ₂
Water gas	Carbon monoxide + Hydrogen	CO + H ₂
Phosgene	Carbonyl chloride	COCl ₂

COMMON TRAP

Plaster of Paris vs Gypsum — a classic trap: Gypsum is CaSO₄·2H₂O (two water molecules). POP is CaSO₄·½H₂O (half water). POP is made from gypsum by heating to 120–130 °C (loses 1.5 of 2 water molecules). When POP is mixed with water, it absorbs water and sets hard — converting back to gypsum. This is used in surgical casts, dental moulds, and sculptures.

PART V — ALLOYS AND METALLURGY

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



The Alloys Table — Priority Memorisation

Alloy composition questions appear in every SSC and UPSC paper. Learn the top 15 cold.

Alloy	Composition	Primary Use
Brass	Copper (60–80%) + Zinc	Utensils, decorative items, musical instruments
Bronze	Copper (~90%) + Tin (~10%)	Statues, coins, bells, ancient tools
Bell metal	Copper (78%) + Tin (22%)	Temple bells
Gun metal	Cu (88%) + Sn (10%) + Zn (2%)	Gun barrels, marine fittings
German silver	Cu (60%) + Zn (20%) + Ni (20%)	Tableware, resistance wires
Stainless steel	Fe (74%) + Cr (18%) + Ni (8%)	Kitchen utensils, surgical instruments
Mild steel	Fe + C (0.1–0.3%)	Construction, bridges
High-carbon steel	Fe + C (0.5–1.5%)	Cutting tools, blades
Cast iron	Fe + C (~3%)	Engine blocks, pipes
Wrought iron	Fe + C (<0.1%)	Ornamental gates, chains
Solder	Pb (50%) + Sn (50%)	Joining electrical wires
Duralumin	Al (95%) + Cu (4%) + Mg + Mn	Aircraft frames, lightweight structures
Magnalium	Al (95%) + Mg (5%)	Balance beams, optical instruments
Nichrome	Ni + Cr + Fe	Heating coil (high electrical resistance)
Alnico	Al + Ni + Co + Fe	Strong permanent magnets
Amalgam	Mercury + another metal	Dental fillings (Hg + Ag + Sn)
Invar	Fe + Ni (36%)	Precision measuring instruments (low thermal expansion)
22-carat gold	Gold 91.6% + Cu/Ag	Jewellery
18-carat gold	Gold 75% + Cu/Ag	Jewellery
Sterling silver	Silver 92.5% + Cu 7.5%	Jewellery, coins

KEY FACT

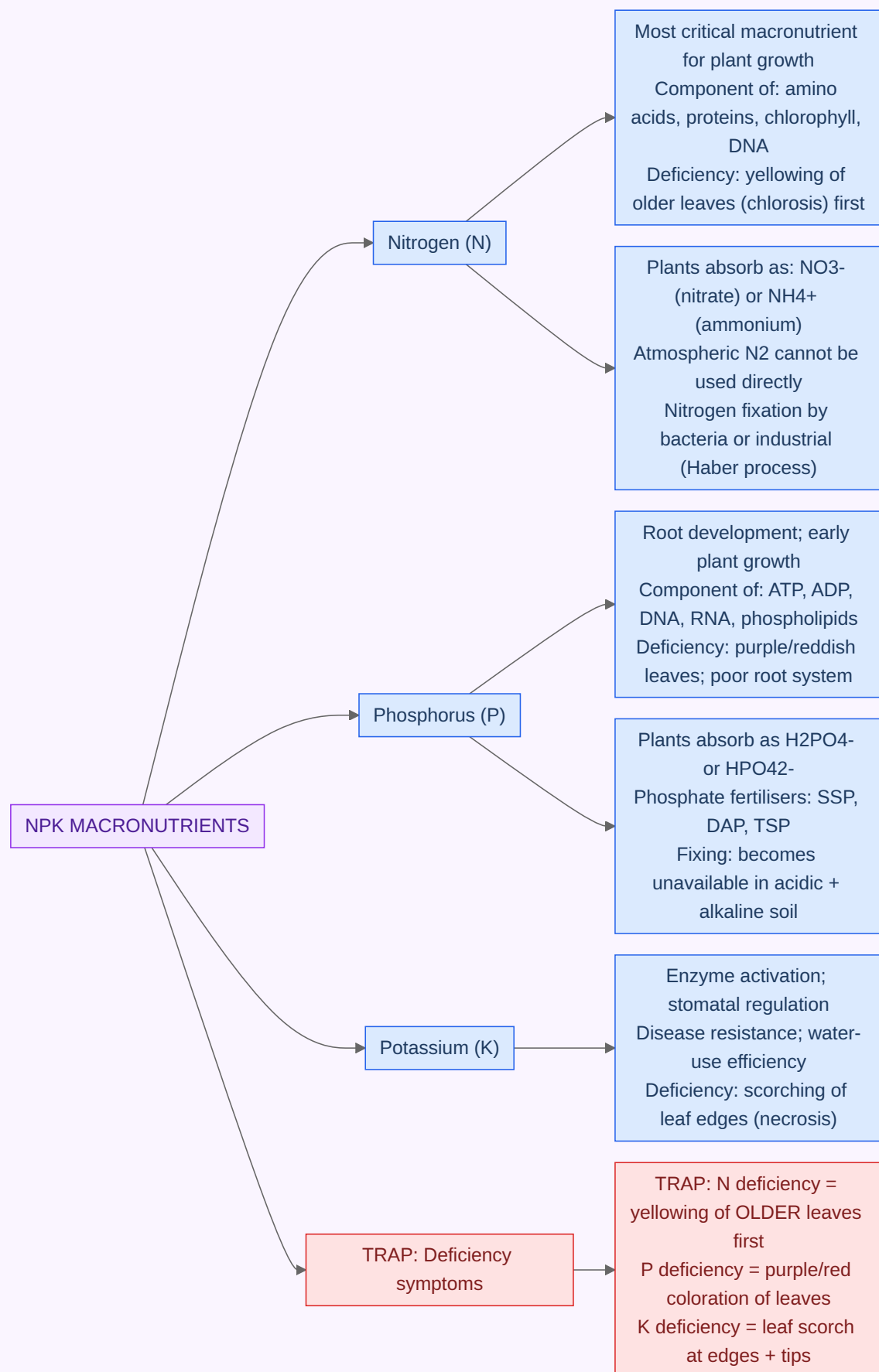
Brass = copper + zinc. Bronze = copper + tin. These two are confused in nearly every exam. Remember: **Brass = Bronze minus Tin plus Zinc**, or just: **Brass = BZ (Bronze with Zinc substituted)**. Bronze has tin; brass has zinc.

PART Y — FERTILISERS

Chapter Y1 — The NPK Macronutrients

Nutrient	Symbol	Role in Plant	Deficiency
Nitrogen	N	Leaf growth, chlorophyll production	Yellowing of older leaves (chlorosis)
Phosphorus	P	Root growth, flower and fruit development	Purple/reddish leaves, poor root system
Potassium	K	Water regulation, disease resistance, stem strength	Leaf-edge scorching (marginal necrosis)

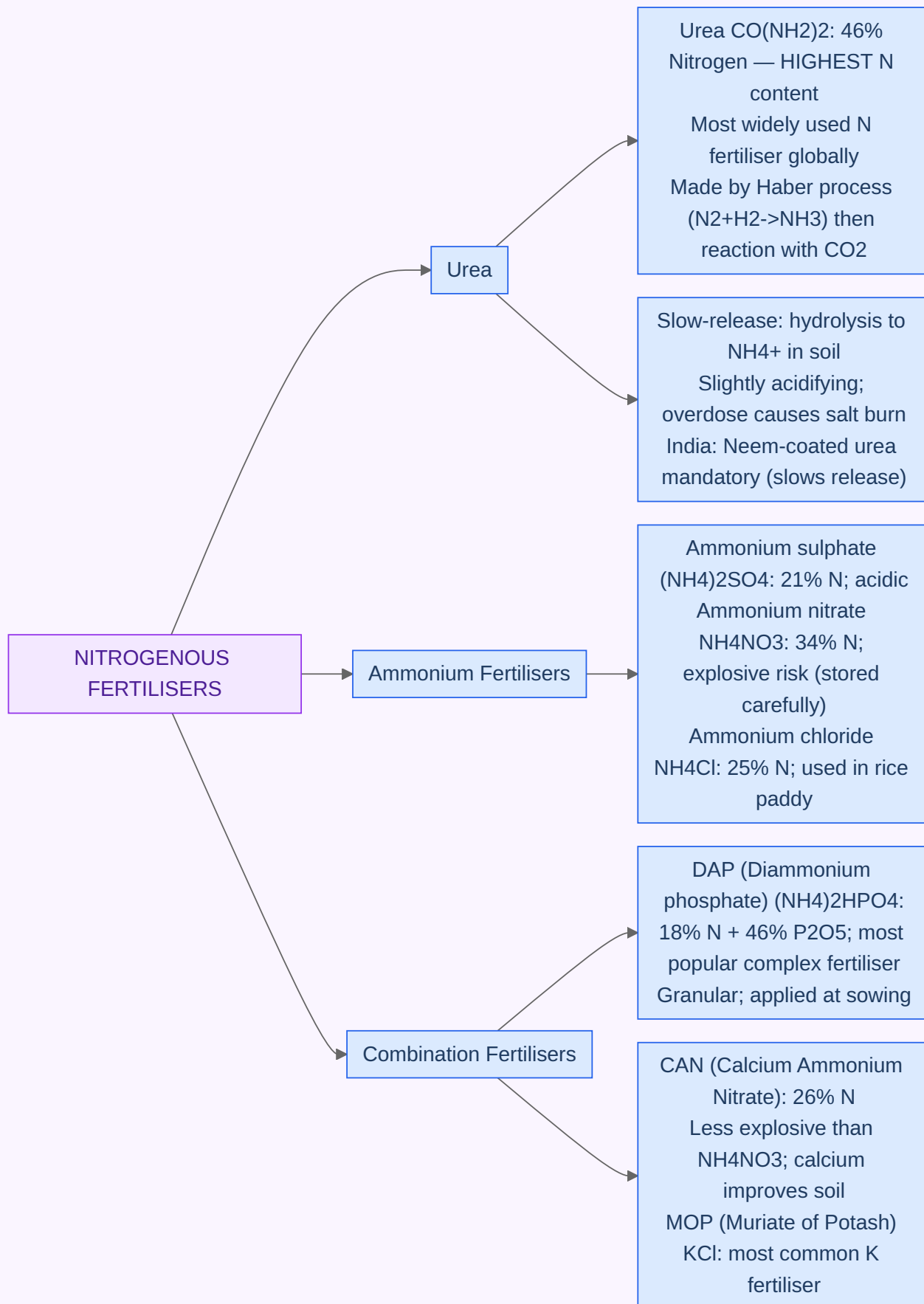
⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter Y2 — Common Nitrogenous Fertilisers

Fertiliser	Formula	% Nitrogen	Key fact
Urea	$\text{CO}(\text{NH}_2)_2$	46%	Highest nitrogen content of all solid fertilisers
Ammonium sulphate	$(\text{NH}_4)_2\text{SO}_4$	21%	Also provides sulphur
Ammonium nitrate	NH_4NO_3	33–34%	Also explosive
Calcium ammonium nitrate (CAN)	$\text{NH}_4\text{NO}_3 + \text{CaCO}_3$	25–26%	Safer form
Sodium nitrate (Chile saltpetre)	NaNO_3	16%	Natural source

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter Y3 — Biofertilisers

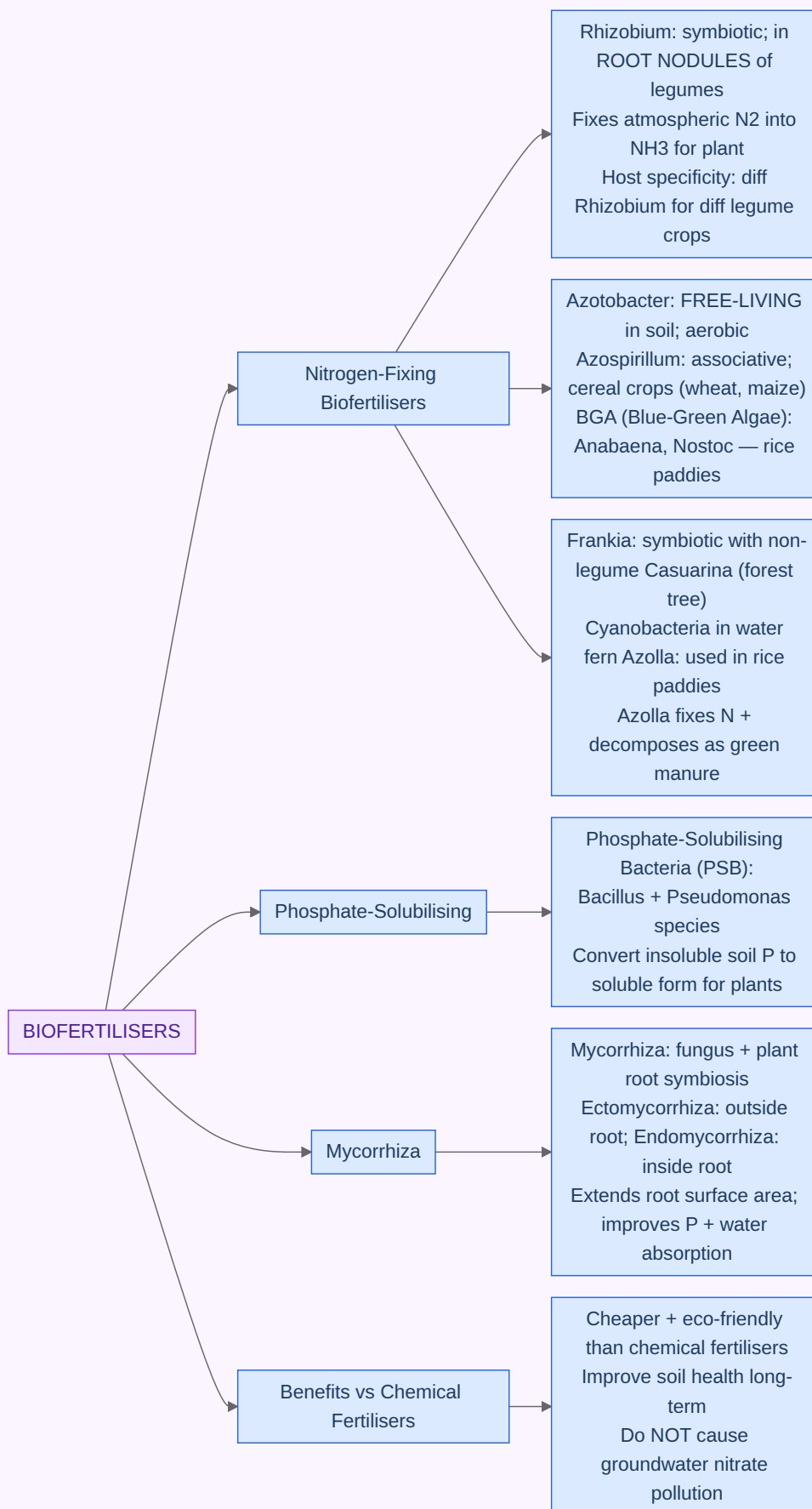
Organism	Type	Action
Rhizobium	Symbiotic (in legume root nodules)	Fixes atmospheric N_2 to NH_3
Azotobacter	Free-living in soil	Fixes atmospheric nitrogen
Azospirillum	Free-living	Nitrogen fixation; plant growth hormones
Anabaena, Nostoc (cyanobacteria)	Free-living	Nitrogen fixation (important in paddy fields)
Mycorrhiza (VAM fungi)	Symbiotic	Extends root uptake area; phosphate mobilisation
Pseudomonas, Bacillus	Free-living	Solubilise insoluble phosphate in soil

EXAM ALERT

Rhizobium bacteria live in root nodules of legumes (pulses — peas, beans, lentils, groundnuts). They convert atmospheric N_2 into ammonia (NH_3) through biological nitrogen fixation. This is why farmers rotate crops — growing legumes restores nitrogen to the soil without chemical fertilisers.

PART AA — INDUSTRIAL PROCESSES

⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



The Process Table — Every Exam Uses This

Process	Product	Key conditions / catalyst
Haber process	Ammonia (NH ₃)	N ₂ + 3H ₂ → 2NH ₃ ; Fe catalyst; 200 atm pressure; 450 °C
Ostwald process	Nitric acid (HNO ₃)	Ammonia + air → NO → NO ₂ → HNO ₃ ; Pt-Rh catalyst
Contact process	Sulphuric acid (H ₂ SO ₄)	SO ₂ + O ₂ → SO ₃ → H ₂ SO ₄ ; V ₂ O ₅ catalyst; 450–550 °C
Solvay process	Washing soda (Na ₂ CO ₃)	NaCl + NH ₃ + CO ₂ + H ₂ O
Castner - Kellner	Caustic soda (NaOH)	Electrolysis of NaCl brine
Hall - Héroult	Aluminium (Al)	Electrolysis of Al ₂ O ₃ in molten cryolite; 950 °C
Bayer process	Alumina (Al ₂ O ₃)	Bauxite + NaOH (leaching)
Bessemer process	Steel from pig iron	Oxygen blast through molten iron
Frasch process	Elemental sulphur	Superheated water + compressed air
Down's process	Sodium	Electrolysis of molten NaCl
Mond process	Pure nickel	Ni + CO → Ni(CO) ₄ → heated → pure Ni
Zone refining	Ultra - pure semiconductors	Si, Ge, Ga — for electronics
Cracking	Petrol from heavier fractions	Thermal or catalytic; breaks long hydrocarbon chains
Reforming	High - octane petrol	Rearranges hydrocarbon molecules; Pt catalyst

REMEMBER

For the three big acid/base industrial processes: "**Haber makes ammonia (H for Hydrogen), Ostwald makes nitric (Ozone-killing NO_x), Contact makes sulphuric (C for Caustic)**". Catalyst clue: Haber = **Fe**; Contact = **V₂O₅** (Vanadium); Ostwald = **Pt** (Platinum).

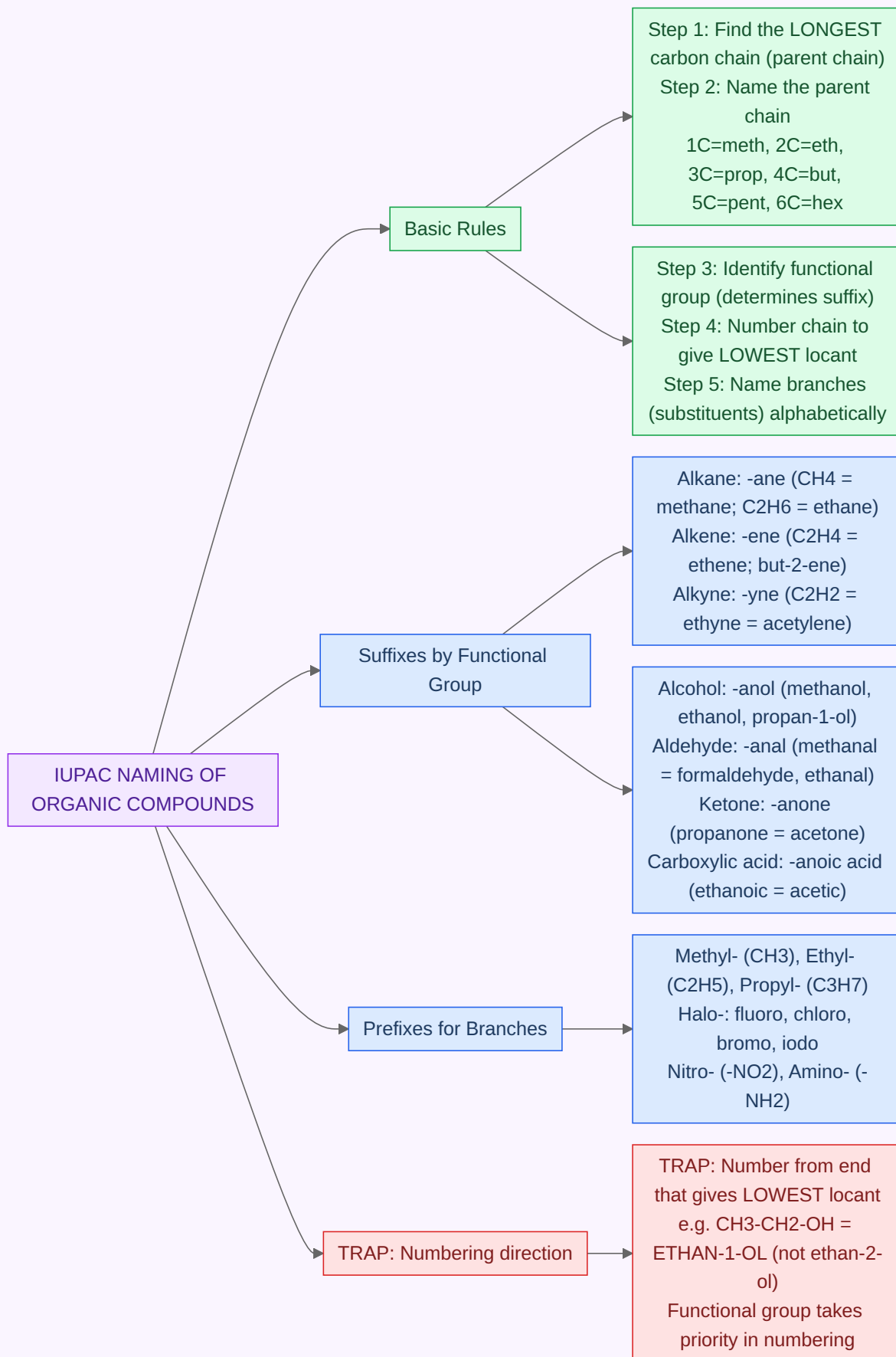
PART AB — ORGANIC CHEMISTRY DEPTH

Chapter AB1 — IUPAC Naming Rules

1. Find the longest carbon chain containing the principal functional group → root name.
2. Number the chain from the end closer to the principal group.
3. Name substituents as prefixes (alphabetical order).
4. Principal group suffix (highest priority first in table below).

Functional Group	Suffix	Example
Carboxylic acid ($-\text{COOH}$)	-oic acid	Ethanoic acid (acetic acid)
Ester ($-\text{COO}-$)	-oate	Ethyl ethanoate
Aldehyde ($-\text{CHO}$)	-al	Ethanal (acetaldehyde)
Ketone ($>\text{C}=\text{O}$)	-one	Propan-2-one (acetone)
Alcohol ($-\text{OH}$)	-ol	Ethanol, methanol
Amine ($-\text{NH}_2$)	-amine	Methylamine
Alkene ($\text{C}=\text{C}$)	-ene	Ethene
Alkyne ($\text{C}\equiv\text{C}$)	-yne	Ethyne (acetylene)
Alkane (no double bond)	-ane	Methane, ethane

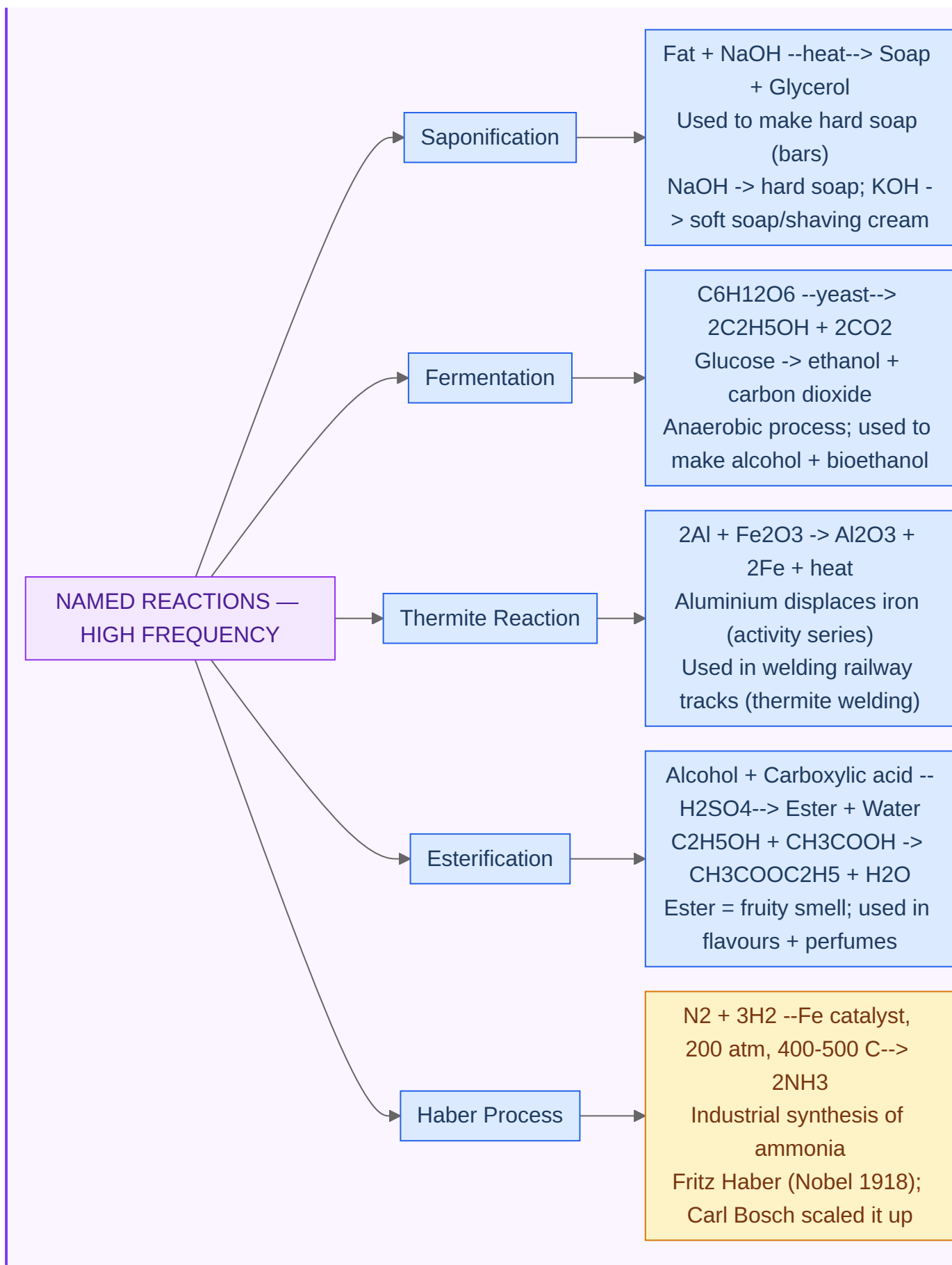
© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter AB2 — Named Reactions (High-Frequency)

Reaction	What happens	Use
Haber	$\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$	Ammonia for fertilisers
Saponification	Fat/oil + NaOH $\rightarrow$ soap + glycerol	Soap making
Esterification	Alcohol + acid $\rightarrow$ ester + water	Perfumes, solvents
Hydrogenation	Alkene + $\text{H}_2 \rightarrow$ alkane (Ni cat)	Vegetable oil $\rightarrow$ Vanaspati ghee
Fermentation	Sugar $\rightarrow$ ethanol + CO_2 (yeast)	Alcoholic beverages, bread
Markovnikov rule	H of HX adds to C with more H atoms	Alkene addition reactions
Wurtz reaction	$2\text{RX} + 2\text{Na} \rightarrow \text{R-R} + 2\text{NaX}$	Synthesise higher alkanes
Aldol condensation	2 aldehydes (with $\alpha\text{-H}$) $\rightarrow$ β -hydroxy aldehyde	C-C bond formation
Friedel-Crafts	Benzene + R-Cl (AlCl_3) $\rightarrow$ alkylbenzene	Aromatic alkylation
Sandmeyer	$\text{Aryl-N}_2^+ + \text{CuX} \rightarrow \text{aryl-X}$	Aryl halide synthesis

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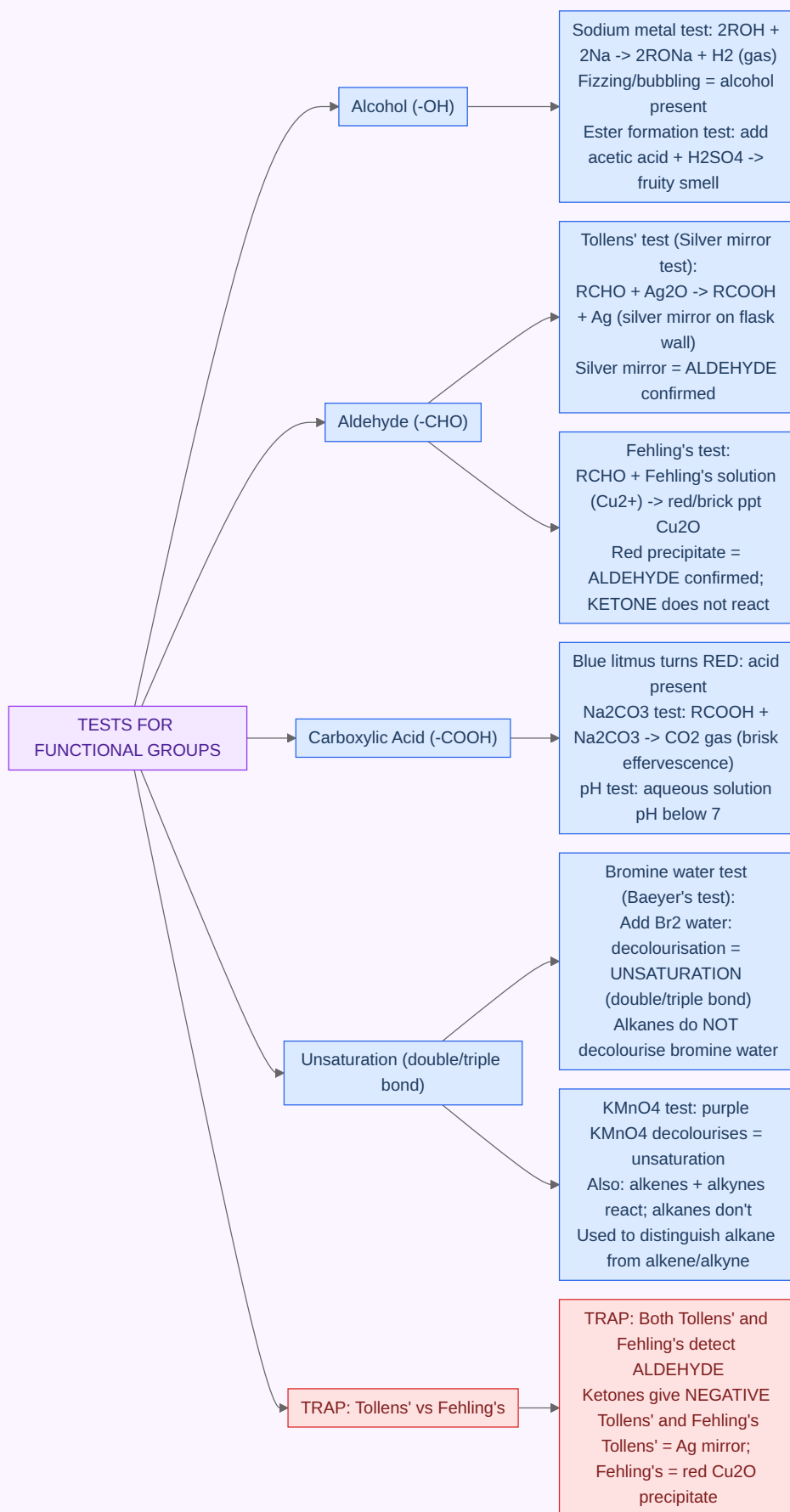


Chapter AB3 — Tests for Functional Groups

Test	What it detects	What you observe
Tollens' (silver mirror)	Aldehyde	Silver mirror forms on test tube wall
Fehling's	Aldehyde / reducing sugar	Brick-red Cu_2O precipitate
Benedict's	Reducing sugar	Red/orange Cu_2O precipitate
Iodoform test	Methyl ketones + ethanol	Yellow CHI_3 (iodoform) precipitate
Lucas test	$1^\circ/2^\circ/3^\circ$ alcohols	Turbidity time: 3° immediate, 2° slow, 1° no reaction
Bromine water	Alkene/alkyne (unsaturation)	Brown colour disappears
Baeyer's (KMnO_4)	Alkene	Purple $\text{KMnO}_4 \rightarrow$ colourless
Sodium bicarbonate	Carboxylic acid	CO_2 effervescence (fizzing)
FeCl_3 (violet)	Phenol	Violet/purple colour
Carbylamine test	Primary amine	Foul-smelling isocyanide

PART AC — BIOMOLECULES

⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Carbohydrates

- Monosaccharides (simple sugars):** Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), fructose, galactose. Cannot be hydrolysed further.

- **Disaccharides:** Two monosaccharides joined. Sucrose = glucose + fructose; Maltose = glucose + glucose; Lactose (milk sugar) = glucose + galactose.
- **Polysaccharides:** Long chains. Starch (α -glucose, plant energy storage), Glycogen (α -glucose, highly branched, animal energy storage in liver/muscles), Cellulose (β -glucose, plant cell walls — humans cannot digest).

EXAM ALERT

Sucrose is NOT a reducing sugar (the functional aldehyde/ketone groups are locked in the glycosidic bond). Maltose, lactose, and glucose ARE reducing sugars. This distinction tests Benedict's/Fehling's test results.

Proteins

- Built from **amino acids** linked by **peptide bonds**.
- 20 amino acids in proteins; 9 are essential (body cannot make them, must come from diet): Histidine, Isoleucine, Leucine, Lysine, Methionine, Phenylalanine, Threonine, Tryptophan, Valine.
- Structure levels: Primary (amino acid sequence) → Secondary (α -helix / β -sheet) → Tertiary (3D folded shape) → Quaternary (multiple subunits, e.g., haemoglobin).

Nucleic Acids

- **DNA** — double helix; deoxyribose sugar; base pairs A-T (2 H-bonds), G-C (3 H-bonds). Carries genetic information.
- **RNA** — single-stranded; ribose sugar; base pairs A-U (uracil replaces thymine), G-C.
- DNA replication is **semi-conservative** — each strand of the double helix is used as a template. Proved by Meselson and Stahl (1958).

Vitamins — The Full Table

Vitamin	Chemical Name	Solubility	Deficiency Disease
A	Retinol	Fat	Night blindness, xerophthalmia
B ₁	Thiamine	Water	Beri-beri
B ₂	Riboflavin	Water	Cheilosis, angular stomatitis
B ₃	Niacin (Nicotinic acid)	Water	Pellagra
B ₅	Pantothenic acid	Water	Burning feet syndrome
B ₆	Pyridoxine	Water	Anaemia, neuritis
B ₇ / H	Biotin	Water	Hair loss, dermatitis
B ₉	Folic acid	Water	Megaloblastic anaemia, neural tube defects
B ₁₂	Cyanocobalamin	Water	Pernicious anaemia
C	Ascorbic acid	Water	Scurvy
D	Calciferol	Fat	Rickets (children), osteomalacia (adults)
E	Tocopherol	Fat	Sterility, RBC fragility
K	Phylloquinone	Fat	Excessive bleeding, poor clotting

REMEMBER

Fat-soluble vitamins: **A, D, E, K** — "Anita Drinks Every Kombucha." All other vitamins (the B family and C) are water-soluble. Only B₁₂ comes exclusively from animal sources — strict vegans need supplements.

APPENDIX A — Master Mnemonics

Activity series (most → least reactive): K, Na, Ca, Mg, Al, Zn, Fe, Sn, Pb, [H], Cu, Hg, Ag, Au, Pt

"Please Send Charlie's Money At Zara For Some Post — He Can't Help Spending All Pounds" (K=Potassium, Na=Sodium, Ca=Calcium, Mg=Magnesium, Al=Aluminium, Zn=Zinc, Fe=Iron/Ferrous, Sn=Tin, Pb=Lead, H=Hydrogen, Cu=Copper, Hg=Mercury/Hydrargyrum, Ag=Silver, Au=Gold/Aurum, Pt=Platinum)

Electronegativity order: F > O > N > Cl > Br > I "French Ocean Navy Cleans British Islands"

Fat-soluble vitamins: A, D, E, K — "Anita Drinks Every Kombucha"

Alkane prefix order (1-10): Meth-, Eth-, Prop-, But-, Pent-, Hex-, Hept-, Oct-, Non-, Dec- "My Elder Parrot Bites People; He's Old Now, Doc"

Periodic table blocks: s (Groups 1-2), p (Groups 13-18), d (Groups 3-12), f (lanthanides + actinides) "smart people do fantastic work" — s, p, d, f

Noble gases: He, Ne, Ar, Kr, Xe, Rn — "He Never Argued; Kept Xerocopy Right"

APPENDIX B — 25-Question Mini-Mock

Attempt without looking at the answers. If you score below 18, revisit the relevant Part.

1. What is the chemical name of bleaching powder?
 2. Aqua regia is a mixture of which two acids and in what ratio?
 3. Which metal is the best conductor of electricity?
 4. What is the chemical formula of washing soda?
 5. Name the process by which aluminium is extracted from bauxite.
 6. What is the monomer of natural rubber?
 7. Name the scientist who discovered penicillin and in what year.
 8. Which vitamin deficiency causes scurvy?
 9. What is the main component of biogas?
 10. In the activity series, which metal is the most reactive?
 11. Brass is an alloy of which two metals?
 12. What is the pH range of human blood?
 13. Which gas turns lime water milky?
 14. What does BOD stand for and what does a high BOD indicate?
 15. Which polymer is used in non-stick cookware?
 16. What is the calorific value of hydrogen compared to petrol?
 17. Name the industrial process that makes ammonia.
 18. Vulcanisation of rubber was discovered by whom and in what year?
 19. Which vitamin is the only one that comes exclusively from animal sources?
 20. What type of bond is present in H₂O that explains its high boiling point?
 21. Which element has the highest electronegativity?
-

22. Name the substance responsible for the "laughing gas" effect.
23. What is the main difference between soap and detergent in hard water?
24. Which disease is caused by mercury poisoning in water?
25. Name three substances that sublime.

Answers: 1. Calcium hypochlorite (CaOCl_2) 2. 3 parts HCl + 1 part HNO_3 3. Silver (Ag) — copper is used in wiring for economy 4. $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ 5. Hall-Héroult process (electrolysis of Al_2O_3 in molten cryolite) 6. Isoprene (2-methylbuta-1,3-diene) 7. Alexander Fleming, 1928 8. Vitamin C (Ascorbic acid) 9. Methane (CH_4 , ~60%) 10. Potassium (K) / Caesium (Cs) 11. Copper + Zinc 12. 7.35 to 7.45 13. Carbon dioxide (CO_2) 14. Biological Oxygen Demand; high BOD = more organic pollution 15. Teflon (PTFE) 16. Hydrogen: ~150,000 kJ/kg vs petrol: ~45,000 kJ/kg (about 3× higher) 17. Haber process 18. Charles Goodyear, 1839 19. Vitamin B₁₂ (Cyanocobalamin) 20. Hydrogen bonding 21. Fluorine (F) 22. Nitrous oxide (N_2O) 23. Soap forms insoluble scum with $\text{Ca}^{2+}/\text{Mg}^{2+}$ ions; detergent works normally in hard water 24. Minamata disease 25. Dry ice (CO_2), iodine, Naphthalene (also camphor, ammonium chloride)

APPENDIX C — Trap Recognition Card

Keep these traps in mind during any chemistry question:

Trap	The wrong answer (popular mistake)	The right answer
Best conductor	Copper	Silver (Ag)
Liquid metal at room temp	Only mercury	Mercury; also gallium (Ga, melts at 29.8 °C)
Lightest metal	Aluminium	Lithium (Li)
First plastic	PVC or nylon	Bakelite (1907)
First synthetic fibre	Polyester	Nylon (1935)
Brass vs Bronze	Both are "copper + something"	Brass = Cu + Zinc; Bronze = Cu + Tin
POP vs Gypsum	Both are calcium sulphate	Gypsum = $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$; POP = $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
GWP strongest gas	CO_2	SF_6 (23,500× CO_2)
Main greenhouse gas	Water vapour	For exams, CO_2 is the expected answer
Ozone protocol	Kyoto	Montreal (1987) targets CFCs
Laughing gas	CO or N_2	N_2O (nitrous oxide)
Mustard gas	Actual gas	Liquid at room temperature — an oily liquid, not a gas
Aqua regia ratio	1:3 or equal	3 HCl : 1 HNO_3 (by volume)
pH of blood	7.0 (neutral)	7.35–7.45 (slightly alkaline)
Sublimation examples	Only dry ice	Dry ice, iodine, naphthalene, camphor